

Chronic hyperactivation of midbrain dopamine neurons causes preferential dopamine neuron degeneration

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This manuscript shows that chronic chemogenetic excitation of dopaminergic neurons in the mouse midbrain results in differential degeneration of axons and somas across distinct regions (SNc vs VTA). These findings are **important** for two reasons. This approach can be used as a mouse model for Parkinson's Disease without the need for the infusion of toxins (e.g. 6-OHDA or MPTP) — this mouse model also has the advantage of showing axon-first degeneration over a time course (2–4 weeks) that is suitable for experimental investigation. Also, the findings that direct excitation of dopaminergic neurons causes differential degeneration sheds light on the mechanisms of dopaminergic neuron selective vulnerability. The evidence that activation of dopaminergic neurons causes degeneration, alters motor behavior, and alters mRNA expression is **convincing**. This is an exciting paper that will have an impact on the Parkinson's Disease field.

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Abstract

Parkinson's disease (PD) is characterized by the death of substantia nigra (SNc) dopamine (DA) neurons, but the pathophysiological mechanisms that precede and drive their death remain unknown. The activity of DA neurons is likely altered in PD, but we understand little about if or how chronic changes in activity may contribute to degeneration. To address this

question, we developed a chemogenetic (DREADD) mouse model to chronically increase DA neuron activity, and confirmed this increase using *ex vivo* electrophysiology. Chronic hyperactivation of DA neurons resulted in prolonged increases in locomotor activity during the light cycle and decreases during the dark cycle, consistent with chronic changes in DA release and circadian disturbances. We also observed early, preferential degeneration of SNc projections, recapitulating the PD hallmarks of selective vulnerability of SNc axons and the comparative resilience of ventral tegmental area axons. This was followed by eventual loss of midbrain DA neurons. Continuous DREADD activation resulted in a sustained increase in baseline calcium levels, supporting a role for increased calcium in the neurodegeneration process. Finally, spatial transcriptomics from DREADD mice examining midbrain DA neurons and striatal targets, and cross-validation with human patient samples, provided insights into potential mechanisms of hyperactivity-induced toxicity and PD. Our results thus reveal the preferential vulnerability of SNc DA neurons to increased neural activity, and support a potential role for increased neural activity in driving degeneration in PD.

Introduction

In Parkinson's disease (PD), the loss of substantia nigra pars compacta (SNc) dopamine (DA) neurons leads to severe disruption of circuit dynamics in the basal ganglia. Compensation for DA loss involves changes in the activity of both surviving SNc neurons, and of other downstream neurons in the circuit. Indeed, following lesions of the nigrostriatal pathway in rats, surviving SNc DA neurons are hyperactive (1), release additional DA (2–5), and have reduced DA reuptake (2). Massive loss of DA neurons (1, 6, 7), complete loss of mitochondrial complex I activity (8), and loss of the mitochondrial PD protein *PINK1* (9) can also result in increased burst firing (10, 11). Therefore, DA neurons are predisposed to altered activity in the setting of extensive loss or stress, which may drive ongoing disease processes. Some lines of evidence support the potential role of hyperactivity in disease initiation, including increased activity in a subset of DA neurons before their degeneration in MitoPark mice (12), increased spontaneous firing in PD patient-derived iPSC dopamine neurons (13), and increased activity of nigrostriatal dopamine neurons in genetic models of PD (9, 14). Moreover, critical PD disease proteins including α -synuclein, LRRK2, *PINK1*, and *Parkin* can influence the level of neural activity (14–19). In particular, the normal function of α -synuclein is believed to be regulating neural activity (20), further supporting the idea that changes in neural activity may contribute to disease pathophysiology. Circuit level changes may also contribute to adverse DA neuron activity. For instance, evidence from primate models suggests that the subthalamic nucleus, which sends a glutamatergic projection to the SNc, is hyperactive in PD (21). While system-level changes may be compensatory and partially restore DA levels and overall motor function, they may also have adverse consequences.

Healthy SNc DA neurons are believed to have immense energetic requirements due to their pacemaking activity, active calcium pumping, unmyelinated or poorly myelinated fibers (22, 23), and large axonal arbors (24). This large energetic requirement likely accounts for their intrinsic vulnerability to mitochondrial insults, including complex I disruption (8, 25, 26) and impairments in mitochondrial dynamics (27) and turnover (28). Neighboring ventral tegmental area (VTA) neurons are relatively spared in PD (29, 30), and this may be due to lower reliance on calcium oscillations for pacemaking (31), their ability to buffer calcium more effectively than SNc neurons (29), and smaller axon arbors (32, 33). Thus, combined with disease-related stress, the metabolic impact of even mild hyperactivity may trigger or accelerate degeneration in SNc DA neurons. In support of this hypothesis, inhibiting the excitatory input from the STN protects SNc DA neurons from 6-OHDA and MPTP toxicity (34, 35). However, empirical evidence linking chronic changes in neural activity to the degeneration of SNc DA neurons in PD is lacking. Recordings from putative SNc neurons in PD patients displayed twofold

higher burst firing when compared to recordings from healthy rodents and nonhuman primates despite similar mean firing rates, though this data is difficult to interpret without human controls (10 [↗](#), 36 [↗](#)).

Additionally, changes in calcium dynamics during hyperactivity can also drive metabolic stress. SNc DA neurons rely on voltage-gated $\text{Ca}_v1.3$ calcium channels to support pacemaking, and blocking these channels is protective against the toxicity of 6-OHDA and MPTP (37 [↗](#), 38 [↗](#)). While classic excitotoxicity involves cytosolic calcium overload and acute cell death, chronic synaptic hyperactivity results in sublethal stress to mitochondria, promoting calcium dysregulation and dendritic atrophy (39 [↗](#)). Mitochondrial calcium overload has been observed in the setting of LRRK2 mutation and PINK1 deficiency (39 [↗](#)). Therefore, altered activity, metabolic stress and calcium overload may all contribute to DA neuron death.

To understand if chronic hyperactivation of DA neurons is sufficient to cause neurodegeneration, we developed a chemogenetic mouse model. Our results indicate that chronically increasing neural activity in midbrain DA neurons results in alteration of circadian locomotion patterns, and prolonged activation leads to selective degeneration of SNc axons and eventual death of midbrain DA neurons. These changes were accompanied by altered intracellular calcium dynamics and transcriptomic changes consistent with calcium dysregulation, supporting a role for increased neural activity in driving neurodegeneration in PD.

Results

To model a chronic increase in DA neuron activity, as may occur in PD, we used a chemogenetic approach. We first expressed the excitatory DREADD hM3Dq specifically in DA neurons using stereotaxic delivery of Cre-dependent hM3Dq AAV to the SNc and VTA of mice expressing Cre under the dopamine transporter (DAT^{IRES} Cre) promoter. Next, we measured the acute behavioral effects of chemogenetic activation of DA neurons. As locomotor output is strongly tied to nigrostriatal DA function (40 [↗](#)–42 [↗](#)), we used home cage wheel running as an *in vivo* proxy for changes in DA function. Mice were single-housed and locomotion was quantified based on wheel rotation. Two weeks after viral injection, we administered clozapine-n-oxide (CNO) by i.p. injection (0.5mg/kg) and confirmed that mice responded with an acute increase in wheel running as an indicator of successful DREADD expression (Figure S1A). The resulting hM3Dq-expressing DAT^{IRES} Cre mice were then administered either vehicle (2% sucrose, to compensate for CNO's bitter flavor (43 [↗](#))) or CNO (2% sucrose, 300 mg/L CNO) via drinking water ad libitum for two weeks (Figure 1A [↗](#)). This strategy allowed us to chronically activate SNc and VTA DA neurons.

Previous reports indicate that chemogenetic activation of DA neurons leads to increased locomotion (44 [↗](#)), but the effects of chronic, long-term activation are unknown. During the first day of treatment, CNO-treated animals were more active than controls during the dark cycle, and also strongly trended toward more activity during the light cycle (Figure 1B [↗](#), 1C [↗](#)). By day 3 of treatment, dark cycle activity markedly decreased in CNO-treated DREADD-expressing mice, and remained decreased for the rest of the treatment time, whereas activity during the light cycle remained increased (Figure 1B [↗](#), 1C [↗](#), Figure S1B). Decreased wheel usage in CNO-treated animals may reflect a consequence of circadian disruption. The prolonged changes in wheel activity indicate that the behavioral effects of chemogenetic activation persist throughout the two-week testing period. In contrast, when DAT^{IRES} Cre mice were given CNO for four weeks in the absence of the DREADD (CNO alone), there were no significant changes in activity in either light or dark cycles (Figure S1C). These results indicate that CNO alone does not significantly alter mouse activity.

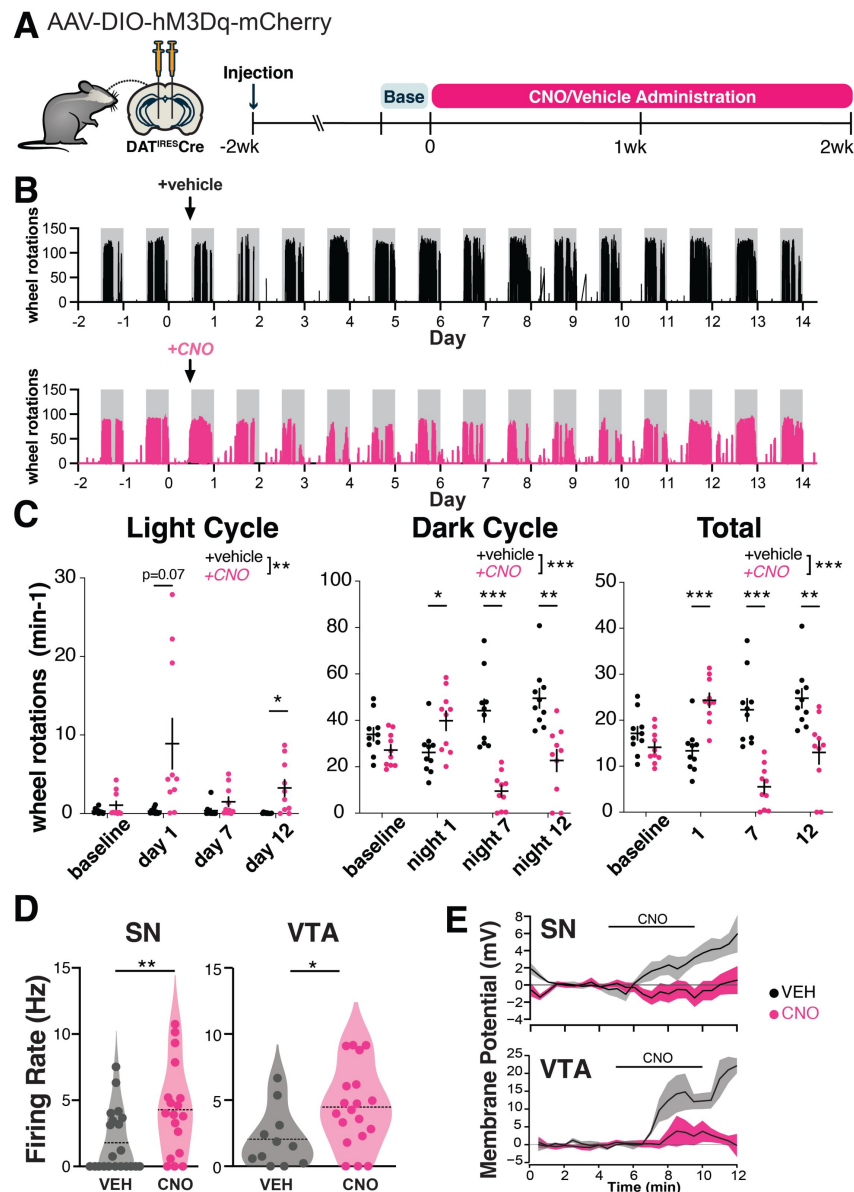


Figure 1.

Chronic hM3Dq activation persistently alters the activity of SNc dopamine neurons.

(A) Graphical illustration summarizing experimental design. Recombinant AAV encoding a conditional allele of the hM3Dq(DREADD)-mCherry was injected bilaterally into the ventral midbrain of 4-5 month-old DAT^{ires}Cre mice. CNO (300 mg/L) or vehicle (2% sucrose in water) was administered *ad libitum* via drinking water for two weeks and the animals were perfused the next day. Changes in locomotion were assessed with running wheels. The two days preceding start of treatment were used as a measure for baseline locomotion.

(B) Representative traces of wheel usage for animals given control vehicle water (top) or CNO water (bottom). Arrows denote start of treatment. Grey background shading indicates dark cycle hours.

(C) Mean wheel usage for selected days during the experiment, segregated by light (left), dark (middle), and total (light + dark; right) cycles. $n=10$ animals/group from 2 independent experiments. * $p \leq 0.05$, ** $p < 0.01$, *** $p < 0.001$ by two-way ANOVA and Holm-Sidak post hoc test.

(D) Spontaneous firing rate was measured during the first 2 min of whole cell recordings. * $p \leq 0.05$, ** $p = 0.01$ by t-test or permutation (non-parametric) analysis.

(E) Time course of responses to bath application of 1 μ M CNO *ex vivo* measured in current clamp in neurons from vehicle-treated and CNO-treated mice.

We performed a similar experiment administering CNO for two weeks to DAT^{IRESc} mice injected with hM3Dq that did not respond to acute CNO IP injection (non-responders; Figure S1D). Non-responders did not show strong initial changes to light or dark cycle wheel usage in the first few days of treatment, but did show persistent decreases in dark cycle usage. Presumably, the lack of early changes reflects either differences in the amount of hM3Dq expressed or the proportion of neurons that are transfected.

To gain insight into the impact of prolonged chemogenetic activation on DA neuron activity, we performed *ex vivo* whole cell recordings in DREADD-expressing SNc neurons in midbrain slices after 7 days of *in vivo* treatment with CNO or vehicle. *In vivo* DREADD activation induced several changes in somatodendritic properties of SNc neurons including a marked decrease in the magnitude of the hyperpolarization activated non-selective cation current I_h (Figure S2A). It also led to an increase in the spontaneous firing rate (Figure 1D, Figure S2C), indicating a persistent increase in firing rate with CNO treatment. The rate of firing was decreased somewhat in controls relative to historical controls from our lab (27), possibly reflecting mild cellular stress from the AAV virus. We did not observe a difference in the coefficient of variation of the interspike interval between treatment groups, thus there was no change in the regularity of pacemaker firing (Figure S2A). *In vivo* CNO led to a more depolarized action potential (AP) peak voltage in spontaneous APs (Figure S2A). CNO treatment did not impact AP threshold voltage or duration (Figure S2A), indicating that AP waveforms were not degraded.

To measure the direct physiological impact of DREADD activation in SNc neurons, we also acutely bath applied 1 μ M CNO to these slices. Interestingly, while SNc neurons from CNO-naïve hM3Dq mice depolarized in response to acute CNO, chronic exposure to CNO for one week *in vivo* eliminated this acute response (in neurons from vehicle-treated mice: 4.9 ± 2.9 mV ($n = 5$), in neurons from CNO-treated mice: -0.5 ± 1.0 mV ($n = 9$); $p = 0.05$, t test, Figure 1E, Figure S2D). Taken together, these findings indicate that 7d of CNO treatment increased spontaneous firing, but altered hM3Dq function such that acute physiological impacts were no longer apparent. The absence of an acute response may indicate a change in the coupling or availability of the receptors, an adaptation or dysfunction within the neurons, or a homeostatic response to prolonged activation. This may also reflect early stages of toxicity to the CNO-treated DA neurons.

We performed similar recordings in DREADD-expressing VTA neurons. As in the SNc, we found that the spontaneous firing rate was higher with 7-days of *in vivo* DREADD activation (Figure 1D, Figure S2C). Unlike the SNc, the coefficient of variation of the interspike interval was decreased in VTA neurons from the CNO-treated mice, indicating more regular firing patterns (Figure S2B). Also unlike the SNc, VTA neurons did not display changes in I_h magnitude or changes in AP peak voltage in spontaneous APs following chronic *in vivo* DREADD activation (Figure S2B). *In vivo* treatment did not impact AP threshold voltage, AP peak, AP duration, input resistance, or initial membrane potential in VTA neurons (Figure S2B).

In response to acute bath application of CNO to slices, VTA neurons from CNO-naïve hM3Dq mice depolarized, while those from mice that received chronic 7-day exposure *in vivo* to CNO showed markedly reduced but not fully eliminated responses (in neurons from vehicle-treated mice: 11.2 ± 8.5 mV ($n = 3$), in neurons from CNO-treated mice: 2.0 ± 2.2 mV ($n = 7$); $p = 0.17$ t test, Figure 1E, Figure S2D), indicating some retained hM3Dq functionality. Together, these data may reflect resilience to chronic activation in VTA neurons compared to SNc neurons.

SNc axons are preferentially vulnerable to chronic hM3Dq-activation

To determine if chronic chemogenetic activation of DA neurons for two weeks induces degeneration, we first quantified its effects on axonal integrity in the striatum. Strikingly, compared to vehicle-treated mice, hM3Dq-expressing animals treated with CNO lost ~40% of their

dopaminergic axons in the dorsal striatum, manifest by decreases in both TH immunoreactivity and reporter mCherry immunofluorescence (**Figure 2A,B**). The same perturbation had a lesser impact on nucleus accumbens and no impact on olfactory tubercle DA afferents. Notably, TH immunoreactivity and mCherry immunofluorescence also decreased in non-responder mice following 2 weeks of treatment (Fig. S3A), although the extent of decrease was somewhat less than in responders. After a shorter 1-week treatment period in a small cohort, we also observed a trend for decreased TH immunoreactivity in the CPu (Figure S3B).

We also assessed the impact of two weeks activation on DA neuron survival in the midbrain. With stereological quantification we found no difference in neuron number between CNO-treated animals and vehicle-treated controls in the SNc or VTA (**Figure 2C**). Therefore, at this time point CNO-treated mice exhibit severe axonal degeneration without DA neuron death.

To ensure that the loss of DA terminals was not due to an off-target effect of CNO, we assessed TH expression in CNO alone animals. These mice did not show a decrease in dopaminergic axons in the caudate putamen (**Figure 2D**), supporting the conclusion that the axonal degeneration requires hM3Dq activation.

We next assessed the impact of more prolonged activation on degeneration of DA neuron somata. Mice treated with CNO for 4 weeks lost the majority of axons projecting to the caudate putamen (**Figure 2E,F**), while fibers in the nucleus accumbens and olfactory tubercle were again preferentially spared. Moreover, following 4 weeks of CNO treatment, there was a significant decrease in the overall number of TH+ midbrain (SNc and VTA) DA neurons, assessed by stereology (**Figure 2G,H**). Although loss of TH reactivity can occur in the absence of neuronal death, there was a similar decrease in the number of mCherry+ DA neurons (labeled by DsRed) in the midbrain. The alignment of these findings supports true loss of DA neurons, although quantitation of mCherry+ DA neurons specifically informs the impact on transduced DA neurons, and this analysis does not account for the variability between injections. Together, these data suggest that, similar to PD progression in humans (45), axonal impairment precedes cell body loss in this model. No preferential loss of SNc versus VTA cell bodies was detected at this time point.

Chronic activation increases baseline population calcium levels in midbrain DA neurons

hM3Dq activation may increase neural activity by increasing intracellular calcium levels (46), and increased neural activity should increase intracellular calcium. To determine the impact of 2 weeks of hM3Dq activation on intracellular calcium concentrations in DA neurons, we bred DAT^{IRESc} mice to GCaMP6f^{fl} mice and injected them bilaterally with the Cre-dependent hM3Dq AAV (**Figure 3A**). The fluorescent signal was used to guide implantation of an optical fiber in the midbrain of animals (**Figure 3B,C**), with confirmed hM3Dq expression based on increased cage wheel running in response to i.p. CNO (Figure S1A). Mice were administered CNO or vehicle water and then placed in arenas for 10-min fiber photometry recording sessions every 1-3 days over the 2-week treatment period. Mean fluorescence levels served as a proxy for population level intracellular calcium concentrations. We observed a small trend for increased fluorescent signal the day after starting CNO that persisted for days 3 through 5 but did not reach significance (**Figure 3D-F**). Interestingly, this was followed by a second, much larger increase in calcium levels between 10-17 days (**Figure 3D-F**), that occurred in parallel with axon loss (**Figure 2A**, Figure S3B). To determine if the increase in neural activity was reversible, we performed two additional recording sessions after removing CNO from the drinking water. Although there was a small trend for decreased fluorescence, this did not reach significance.

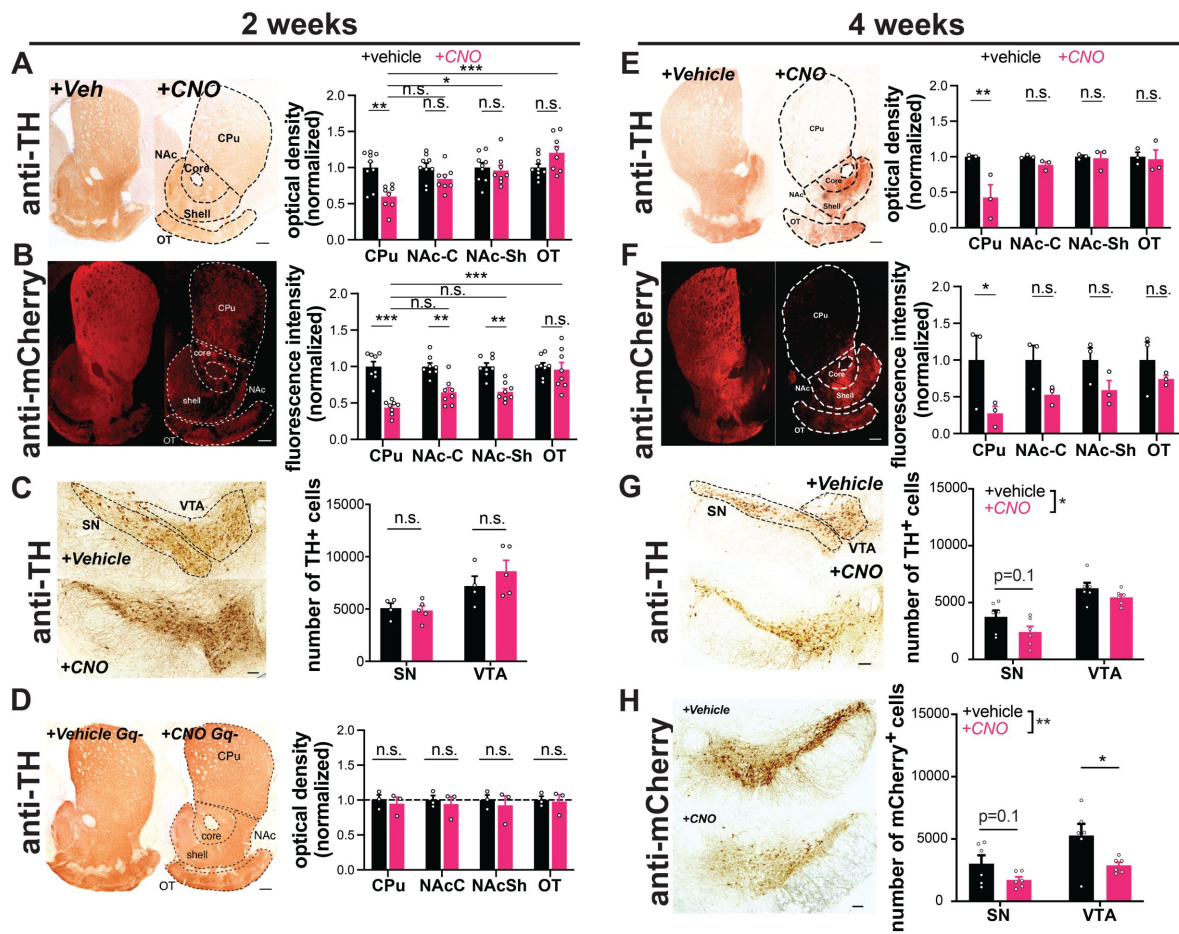


Figure 2.

Chronic activation with AAV-hM3Dq-DREADDs is preferentially toxic to nigrostriatal axons.

DAT^{IRESc} mice expressing hM3Dq(DREADD)-mCherry (A-C, E-H), or no virus injection (D) in DA neurons. Example images of TH (A,D,E) and mCherry (B,F) immunoreactivity in striatal sections of mice treated for two or four weeks with vehicle (left) or CNO (right) via drinking water. DA neuron projection areas in dorsal and ventral striatum are indicated with dotted lines. Quantifications for TH (A,D,E) and mCherry (B,F) optical density at 2 and 4 weeks show preferential loss in CPu. $n = 8-9$ (A), 8 (B), or 3 (D, E, F) animals/group, 3-5 sections/animal. (C,G,H) Images of TH and mCherry (4 weeks only) immunoreactivity in midbrain of vehicle (top) or CNO (bottom) treated mice at 2 or 4 weeks. SN and VTA regions are indicated with dotted lines. (C,G,H) Stereology estimating the number of TH+ or mCherry+ DA neurons. Chronic CNO treatment of hM3Dq(DREADD)-expressing mice shows a significant decrease in both TH and mCherry immunoreactivity and DA neuron number. $n = 4-5$ hemispheres (one per mouse, C) or 6 hemispheres (two per mouse, G, H) per group. Scale bars indicate 100 μ m in the midbrain and 200 μ m in the striatum. Error bars indicate mean $\pm$ SEM. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ by two-way ANOVA and Holm-Sidak *post hoc* test. SN: substantia nigra, VTA: ventral tegmental area, CPu: caudate putamen, NAc-C: nucleus accumbens core, NAc-Sh: nucleus accumbens shell, OT: olfactory tubercle.

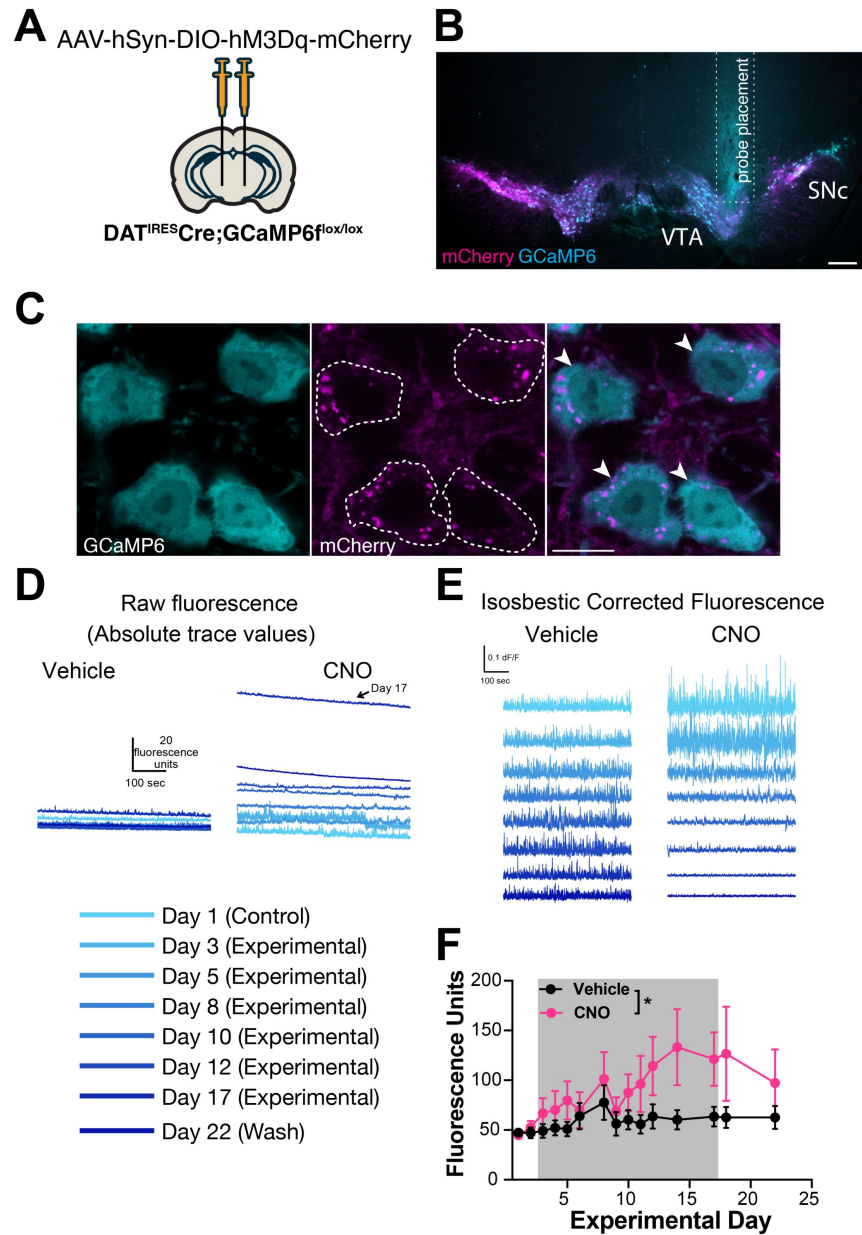


Figure 3.

Chronic hM3Dq activation increases baseline calcium in parallel with axonal degeneration.

(A) Transgenic mice expressing calcium-reporter GCaMP6 specifically in DA neurons were injected bilaterally with AAV-DIO-hM3Dq-mCherry and implanted with an optical probe for baseline calcium measurements during a 14-day chronic chemogenetic activation.

(B) Representative image of photometry probe placement in mouse midbrain to record from dopamine neurons co-expressing hM3Dq-mCherry (magenta) and GCaMP6 (cyan). Scale bar is 200 μ m.

(C) Representative high-magnification images of reporter mCherry (magenta) and GCaMP6 (cyan) co-expression. Scale bar is 10 μ m.

(D) Representative raw traces of baseline calcium fluorescence in mice treated with vehicle vs CNO.

(E) Representative isosbestic corrected traces in mice treated with vehicle vs CNO.

(F) Baseline calcium fluorescence levels of DA neurons in mice treated with vehicle or CNO for 14 days (gray shaded area), and following wash. Error bars indicate mean $\pm$ SEM. *, $p < 0.05$ by two-way ANOVA, $n = 7$ mice/group from 2 independent experiments.

Although GCaMP fluorescence is not quantitative, the consistent difference in baseline in animals tested concurrently on the same system provides confidence in our interpretation. The frequency and amplitude of calcium transients decreased with chronic CNO treatment, and during washout the frequency remained decreased while the amplitude returned to vehicle levels (Figure S4A). Notably, the decrease in frequency or amplitude is unlikely to represent a ceiling effect (i.e. where a high baseline might occlude transient detection), because while the absolute baseline and transient amplitude returned to pre-treatment levels during the washout, the frequency of transients did not. Instead, the decreases in transient frequency and amplitude may reflect less bursting activity in the neurons. As sharp peaks in genetically encoded calcium indicator signals tend to reflect bursting activity, the sustained decrease in frequency during washout may reflect more permanent dysfunction in the burst firing patterns of these neurons.

Together, these data support massive increases in DA neuron intracellular calcium over time, which may be due to hM3Dq receptor activation, calcium dysregulation, and the onset of degeneration. We also assessed time spent moving and distance traveled during photometry sessions in a subset of animals (Figure S4B). Paralleling the small initial increase in baseline calcium, there was a strong trend for CNO to increase activity over time versus controls in both measures of gross locomotion during the first week of treatment (distance traveled $p=0.07$, percent time spent moving $p=0.06$). Interestingly, the increased open field movement in the CNO group decreased back to baseline by day 11 (Figure S4B), just as calcium levels begin to climb markedly (Figure 3F), and in parallel to axonal degeneration (Figure 2A, Figure S3B).

Identifying activation-associated transcriptomic changes with region specificity using spatial transcriptomics

To gain additional insight into the mechanisms of degeneration in this model, we used spatial genomics. We performed Visium spatial transcriptomics on midbrain and striatal sections from DAT^{ires}Cre mice that were bilaterally injected with AAV-DIO-hM3Dq-mCherry at 3 months and then received either CNO (GqCNO) or vehicle (GqVeh) drinking water for one week prior to harvest. Additional non-injected control mice also received CNO (CNO alone) (Figure 4A). Mice were harvested following one week of CNO treatment in order to quantify transcriptomic changes at a time point when some axons still remain and prior to somatic degeneration (Figure 2A, Figure S2A, Figure S5A), thereby focusing on early gene expression changes prior to neuronal death.

We next identified 55 μm diameter barcoded discs (Visium Spatial Gene Expression) overlaying regions enriched for DA neurons within the SN and VTA in the midbrain, and the caudate putamen (CP) (Figure 4B). In the midbrain, selected discs contained at least one entire TH-positive (and mCherry-positive for injected animals) soma, and also expressed at least two of three characteristic DA genes (DAT, vesicular monoamine transporter 2 (VMAT2), or TH) above a pre-set threshold level (Figure 4C). Interestingly, DAT and VMAT2 were significantly decreased in the SN and VTA of GqCNO mice, perhaps indicating a loss of dopaminergic phenotype. In agreement with this, we found that striatal dopamine levels were markedly decreased by HPLC (Figure S6). Indeed, principal components analyses of the SN and VTA in the midbrain demonstrate that GqCNO midbrain dopaminergic regions are transcriptionally similar to one another and distinct from control groups (Figure 4D). This suggests that chronic DREADD activation induces a robust transcriptomic response in DA neurons that is independent of viral transduction or CNO administration. Consistent with this, we employed the NEUROeSTIMator method (47) to predict neural activity scores based on the transcriptomic assessment of the SN and VTA, finding that GqCNO samples had higher predicted activity scores than controls (GqVEH and CNO alone) (Figure 4E).

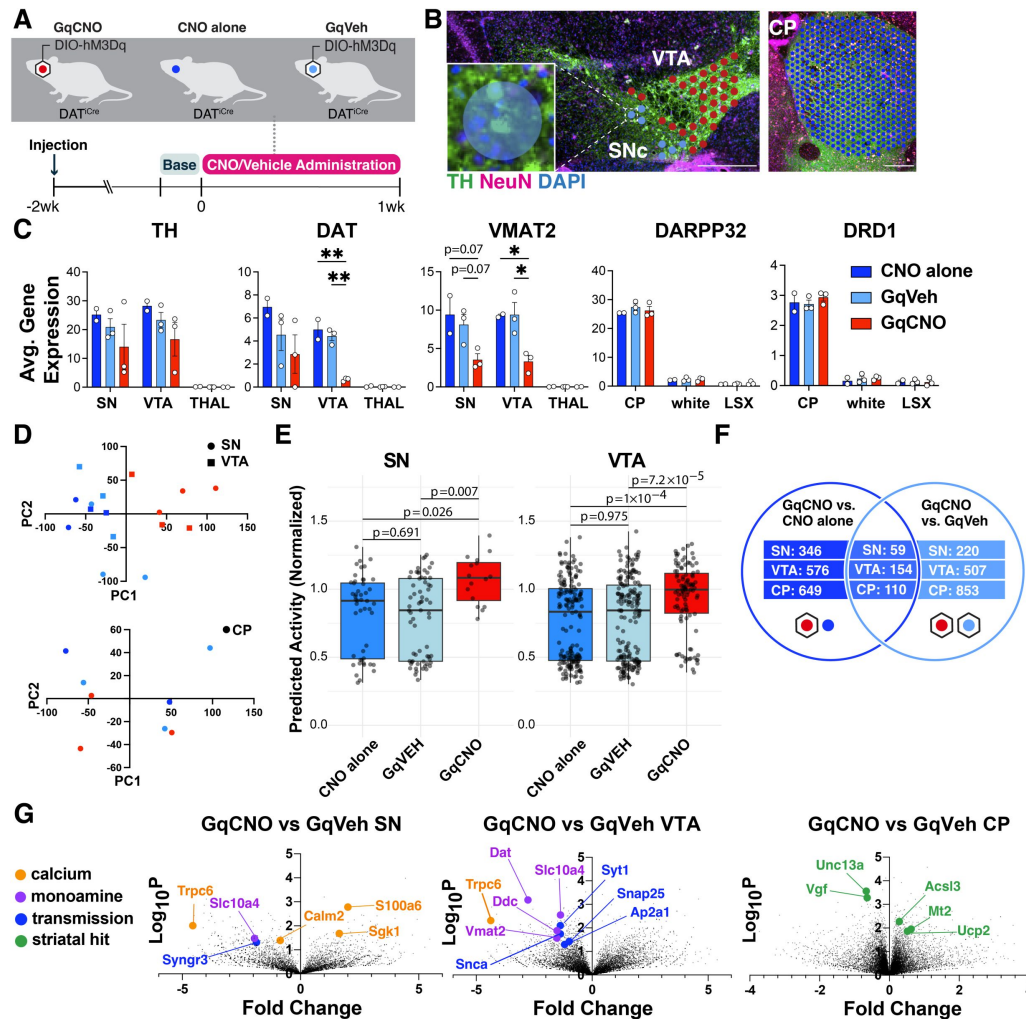


Figure 4.

Spatial transcriptomics reveals midbrain DA and striatal target DEGs altered by chronic DA neuron hyperactivity

(A) DAT^{ires} Cre animals that received CNO (CNO alone, n=2 mice) or were injected with AAV-hM3Dq-mCherry and received vehicle (GqVeh, n=3 mice) or CNO (GqCNO, n=3 mice) were treated for one week before brains were flash frozen for spatial transcriptomic analysis.

(B) Image of midbrain and striatal sections stained with TH (green), NeuN (purple), and DAPI (blue) shows discs assigned to regions of interest. Inset shows a disc containing two TH+ cell bodies. Scale bars indicate 500µm.

(C) Expression of dopaminergic and striatal genes is confined to expected spatial regions. Expression of genes involved in DA metabolism decrease with chronic CNO. 2-49 discs were compiled per VTA and 1-7 discs were compiled per SN. 357-560 capture areas were compiled per CP. The thalamus was selected as a midbrain control region, while white matter tracts (white) and the lateral septal complex (LSX) were used as striatal controls. *p<0.05, **p<0.01, ***p<0.001 by one-way ANOVA followed by Holm-Sidak *post hoc* test in the SN, VTA, and CP.

(D) Principal components analysis of midbrain regions (top) and the caudate putamen (bottom) for GqCNO, GqVeh, and CNO alone groups.

(E) The deep learning NEUROeSTIMator model was used to predict neural activity of GqCNO, GqVeh, and CNO alone within the SN and VTA from Visium spatial transcriptomics. Groups were compared using KS-test (see Methods).

(F) Hits were used for Enrichr pathway analysis if significant in both GqCNO vs CNO alone and GqCNO vs GqVeh comparisons. Gene rankings for hit analysis were established using fold change score (FCS) and signal-to-noise score (SNS).

(G) Volcano plots comparing GqCNO vs GqVeh in the SN, VTA, and CP. Genes highlighted are also significantly altered when comparing GqCNO vs CNO alone.

We hypothesized that chronic activation of DA neurons likely also leads to specific gene expression changes in striatal target neurons. In the striatum, discs were selected based on expression of *DARPP32* (**Figure 4C**). Notably, principal components analysis between GqCNO and control groups revealed less separation in the striatum than the midbrain (**Figure 4D**), perhaps because of the multiple inputs on striatal neurons in addition to dopaminergic axons, although future investigation is required.

We next compared the transcriptomic profiles of discs within each brain region between GqCNO and GqVeh mice, as well as in GqCNO vs CNO alone mice. Only genes that had significant differential expression between GqCNO and both control groups were assessed for pathway changes (**Figure 4F**). Consistent with chronic DREADD activation, pathway analyses of GqCNO vs control groups in the SN and VTA showed significant enrichment of GO Biological processes including “chemical synaptic transmission”, “synaptic vesicle exocytosis”, “positive regulation of transporter activity”, and “vesicle transport along microtubule” (Table S1). Also consistent with DREADD activation, GO Molecular Functions in the VTA included “GTP binding” and “syntaxin binding” (Table S1). Interestingly, both CP and SN GO Molecular Functions included “calcium channel regulator activity”, suggesting a larger circuit-level change in calcium channel regulation as a result of chronic DREADD activation (Table S1).

Consistent with this, there were a number of differentially expressed genes in the SN and/or VTA that regulate neural transmission, calcium, and activity (**Figure 4G**, Figure S5B, S5C, Figure S7, Figure S8). These included increased expression of *S100a6*, a calcium sensor and modulator (48), and *Sgk1*, a serine/threonine kinase regulated by intracellular calcium involved in the regulation of ion channels, membrane transporters, cellular enzymes, and transcription factors (49). Interestingly, *Sgk1* expression trended up in the CP, supporting circuit-level changes in calcium signaling. Increased *S100a6* expression has been observed in AD and ALS (50), while *Sgk1* upregulation has been associated with DA neuron death in rodent toxin models of PD (51, 52). Conversely, the expression of calmodulin-2 (*Calm2*), which mediates control over a large number of enzymes and ion channels as a result of calcium binding (53), and *Trpc6*, which is thought to form a non-selective calcium-permeant ion channel (54) were both decreased. Decreased *Trpc6* may be a homeostatic response to chronically high intracellular calcium. These changes in calcium-regulating genes are consistent with the changes in baseline intracellular calcium observed in DA neurons by GCaMP photometry (**Figure 3**), and may reflect early processes that either cause or contribute to the onset of degeneration, or alternatively compensate to protect against degeneration.

Pathway-level changes in transmission were driven by decreases in several synaptic-vesicle genes, including: synaptotagmin 1, a calcium sensor that triggers transmitter release including striatal dopamine release (55); Snap25, a t-SNARE that regulates transmitter release (56); Ap2a1, a component of the adapter protein complex AP-2 (57); and Syngr3, a synaptic vesicle-associated protein that may regulate DAT function (58). Interestingly, α -synuclein was also significantly downregulated in the VTA. α -synuclein is enriched in presynaptic terminals, and has been shown to modulate DA release (59). Together, these results suggest that chronic DREADD-activation of DA neurons results in significant changes in synaptic transmission which may result in altered calcium regulation and DA release.

To begin to interrogate the role of calcium in hyperactivity-induced degeneration, we performed a pilot experiment in which mice injected with saline in the left midbrain and AAV-hM3Dq(DREADD)-mCherry in the right midbrain were then implanted subcutaneously with slow-releasing placebo or isradipine pellets (Figure S9A). Mice were not acutely tested for responsiveness to CNO, as wheel running data is less clear with unilateral manipulation. Instead, DREADD expression was validated *post hoc* by immunofluorescence, and animals with little or no DREADD expression were excluded from further analysis. Isradipine blocks $\text{Ca}_v1.2$ and $\text{Ca}_v1.3$ channels that are highly expressed in SNc neurons (60, 61). Isradipine treatment has been

shown to return SNc DA neurons to a juvenile form of pacemaking, and chronic treatment has been shown to reduce calcium-dependent mitochondrial oxidant stress (60, 62). To confirm the presence of systemic isradipine, we measured isradipine levels in blood plasma via HPLC (Figure S9B). Following two weeks of vehicle or CNO administration with placebo or isradipine release, we assessed striatal TH immunoreactivity and mCherry immunofluorescence. Consistent with previous results, we observed decreases in TH and mCherry in CNO-treated AAV-Gq hemispheres, but chronic isradipine administration did not rescue this loss (Figure S9C, S9D). Notably, the absence of rescue may result from a lack of statistical power due to small sample size, or due to the presence of non-responders. Interestingly, chronic isradipine induced some decrease in TH immunoreactivity in vehicle-treated AAV-Gq hemispheres without reductions in mCherry, suggesting that chronic isradipine may induce TH downregulation without degeneration.

We also identified differentially expressed genes (DEGs) in the striatum. Upregulated genes included: Mt2, a metallothionein protein that acts as an anti-oxidant to protect against free radicals (63); Acsl3, an acyl-CoA synthetase that stimulates fatty acid synthesis (64) and could supply energy substrates; and the mitochondrial uncoupling protein Ucp2. Downregulated genes included: Unc13a, which is involved in neurotransmitter release via vesicle maturation and has been implicated in ALS (65, 66), and Vgf, a secreted growth factor whose expression is also decreased in PD (67).

We also evaluated the SN DEGs (n=59) and additional genes of interest identified within our DREADD mouse model with gene expression changes within human control and early PD subjects (Figure S10A,B). We employed the Nanostring GeoMx assay to assess whole transcriptome changes of dopamine neurons (TH+ masked, see Methods) within the SNc ventral tier of Controls and Early PD subjects. Of the 59 DEGs, 51 were found within the human dataset, and these genes exhibited broadly congruent directional changes between mouse and human (Chi-square p-value = 0.05). This suggests that similar mechanisms may be at play in the PD midbrain and the DREADD mouse model. To investigate this further, we considered genes that emerged from the mouse dataset involved in calcium, transmission, and dopamine, and evaluated them in the early PD samples (Figure S10B, S10C). We found that expression of Syt1, Syng3, and Calm2 were all decreased in both mouse and human datasets. Similarly, we found that expression of Slc18a2 (VMAT2) and Slc10a4 were also both decreased in the mouse and human data. These findings may suggest common mechanisms of calcium dysregulation and altered dopamine metabolism in an early stage of degeneration in our mouse model and in human PD. Additionally, differential gene expression analysis revealed that HSPA4 and EIF3G were downregulated (FDR < 0.02) within the SNc ventral tier DA neurons of early PD subjects compared to controls and in the mouse model (Figure S10A) highlighting other potential common pathogenic mechanisms. Additional investigation is required to determine the functional impact of these genetic changes on dopamine neuron and dopamine neuron subtype function and vulnerability.

To further assess these findings on a single cell basis, we performed high resolution Xenium spatial transcriptomics on adjacent sections from the same tissues (Figures S11). Sections processed for Xenium were then processed for mCherry immunofluorescence. Expression of mCherry, along with anatomy, and expression of characteristic dopaminergic genes (TH, DAT, VMAT2) were used to select AAV-transduced dopaminergic nuclei in the SNc and VTA. In these nuclei, TH and VMAT2 expression levels were similar across groups, but expression of DAT was decreased in GqCNO nuclei. Similar to the lower-resolution Visium, expression of Slc10a4 was also decreased in GqCNO nuclei. Interestingly, other markers of vulnerable SNc neurons were also decreased in GqCNO SNc nuclei, including Aldh1a1, Sox6, and Anxa1 (68–70). Downregulation of Aldh1a1 has also been reported in PD (71). These subtype markers cannot be assessed in the lower-resolution Visium dataset due to their relative scarcity and the large size of the Visium capture areas, but their differences are appreciable with the single-nucleus resolution of Xenium. Together, these results support a loss of a dopaminergic phenotype, especially in vulnerable nigral dopaminergic nuclei.

Discussion

Here we establish a new model system to chronically activate midbrain DA neurons, and show that this prolonged activation leads to the preferential and step-wise degeneration of SNc DA neurons, starting with loss of terminals and progressing to neuronal death.

New model to chronically activate DA neurons

Prior studies have activated midbrain DA neurons by both optogenetics and DREADDS to study the impact of DA neuron activity on sleep (72 [↗](#)) and the spread of pathological α -synuclein (73 [↗](#)). However, in order to accurately model the impact of neural activity on neurodegeneration, we hypothesized that activity must be chronically increased. Unlike other studies which have used once or twice daily injections of CNO, we delivered CNO in the drinking water to ensure a more continuous, sustained impact on neural activity. We confirmed that chronic 1, 2, or 4 week hM3Dq activation results in sustained increases in intracellular calcium levels and functional behavioral readouts, suggesting that our model induces long-lasting changes in dopamine neuron activity. Moreover, *ex vivo* electrophysiology demonstrated that the basal level of DA neuron activity does indeed increase with CNO treatment. An important question, however, is whether our model accurately recapitulates how neural activity increases in physiologic or pathologic situations.

Compensatory increased or decreased rates of DA neuron pacemaking may help to maintain striatal DA as it is depleted in PD, or to protect against energy failure by decreasing energy consumption. Meanwhile, changes in bursting may reflect circuit-level adaptations to PD in response to ongoing degeneration. Irregular or silenced pacemaking is commonly observed in toxin-based models of PD (74 [↗](#)–78 [↗](#)), while increased induced bursting is observed after massive loss of DA neurons or complete loss of complex I (1 [↗](#), 6 [↗](#)–8 [↗](#)). These changes could further promote the degeneration of remaining DA neurons. As such, in future studies it will be important to understand how changes in the pattern of activity (i.e. bursting versus pacing) impact degeneration. In other PD models, DA neuron activity is altered prior to degeneration (9 [↗](#), 12 [↗](#), 13 [↗](#)). The energetic demands of neural activity position altered activity as a possible contributor to the degenerative process in energetically vulnerable DA neurons, either as an initiating force or as a driver of disease progression.

One possible outcome of chronic hyperactivation is depolarization block (DB), a process that occurs even in healthy neurons wherein prolonged excitation induces persistent depolarization and blockade of spike generation (79 [↗](#)). DB occurs in midbrain DA neurons following chronic treatment with antipsychotic drugs (80 [↗](#)), and DB in SNc neurons specifically is associated with extrapyramidal symptoms (81 [↗](#), 82 [↗](#)). Lesions of DA neurons with 6-OHDA can also increase susceptibility to DB (83 [↗](#)). In our chronic hyperactivation model there is a strong decrease in dark cycle wheel running behavior that may be consistent with DB (Figure 1C [↗](#), Figure S1B). Prolonged excitation along with progressive degeneration in our model may put DA neurons at greater risk of entering DB. However, CNO-treated animals have increased rather than decreased open field movement (Figure S4B). Our NEUROeSTIMator data also predicts higher activity scores in GqCNO samples compared to GqVEH and CNO alone controls (Figure 4E [↗](#)). Moreover, while DB can't be assessed in slices due to its reliance on afferent input, we observed increased pacemaking in CNO-treated neurons, further arguing against prominent induction of DB. Given that VTA and SNc neurons can have differential susceptibilities to DB in response to antipsychotics, it is possible that a subpopulation of DA neurons in our model periodically enter DB without gross effects on locomotion. Additional data such as *in vivo* electrophysiology will be necessary to fully investigate the possibility of DB.

As expected, chronically increasing neural activity increased the movement of mice during the light cycle. However, it was notable that movement as measured by wheel use in the home cage did not increase as robustly as did movement in the open field chamber (Figure S4B), perhaps indicating that the impact on movement is greater in novel environments. This may reflect the effects of midbrain DA neurons on motivation and exploration (44, 84). The later return to control levels of locomotion midway through CNO treatment may indicate the onset of synaptic dysfunction or degeneration. In addition, our electrophysiologic analyses focused on cell body function, but the loss of terminals starting by ≈ 1 week (Figure S3B) raises the possibility that synaptic function might be disrupted even earlier, in parallel to behavioral changes. Meanwhile, chronically increasing activity unexpectedly and robustly decreased movement during the dark cycle (when mice should be most active) within 3–4 days of starting CNO (Figure 1C, Figure S1B). One possibility is decreased dark cycle movement results from a disruption in sleep, which might occur given the mild increased use of running wheels during the light cycle. Indeed, prior studies show that acute inhibition or stimulation of VTA DA neurons by optogenetics or DREADDs suppresses or promotes wakefulness, respectively (72, 85). These data support a critical role for VTA DA neurons in maintaining wakefulness, and suggest that increased activity might promote wakefulness in our activity model. Future investigations assessing the impact of chronic SNc versus VTA DA neuron activation on circadian rhythms and sleep are needed to determine if these DA neurons similarly influence sleep in the setting of prolonged stimulation, and how potential sleep changes might contribute to the effects of chronically increased neural activity on motor function.

Chronic chemogenetic activation drives degeneration

How does increasing the activity of SNc and VTA DA neurons impact these neurons, as well as their striatal targets? Analysis by spatial transcriptomics near the onset of axonal degeneration reveals a clear decrease in the expression of genes involved in DA synthesis (TH), uptake (DAT) and storage (VMAT2) in the SNc and VTA, perhaps reflecting neuronal stress that precedes degeneration or an attempt to decrease DA release. Further investigation will be needed to determine whether these gene expression changes are associated with functional deficits. Decreases in presynaptic transmission-related genes (Syt1, Snap25, Snca, Ap2a1) in midbrain dopamine neurons are consistent with axonal dysfunction and loss that are observed later in CNO treatment. Moreover, consistent with the role of hM3Dq DREADDs in increasing cytosolic calcium (46) on a systems level, we observed a significant change in expression of genes involved in the regulation of calcium channels (Supplemental Table 1), as well as serum and glucocorticoid-inducible kinase 1 (Sgk1), a protein-serine/threonine kinase activated by calcium (86), in both the midbrain and striatum. Our photometry analysis also raise the possibility of a central role for calcium in activity-induced degeneration, as calcium levels in midbrain DA levels rose markedly following 7 days of CNO (Figure 3F), as axonal degeneration progresses (Figure S3B).

The mechanism of preferential degeneration remains to be determined, but is likely triggered by increased cytosolic calcium levels associated with hM3Dq activation (46). SNc DA neurons are believed to be intrinsically vulnerable to increased calcium due to their pacemaking activity driven by $Ca_v1.3$ channels (60), combined with the high energetic requirements of removing calcium from neurons and lack of calbindin expression (33). Further investigation is required to determine if DREADDs differentially increase calcium levels in SNc vs VTA DA neurons, and/or if SNc DA neurons are more vulnerable to the same level of elevation. Further, it remains unknown whether increased calcium leads directly to degeneration, for instance by triggering energy failure or increasing ROS, or if it indirectly causes neurodegeneration by triggering potentially toxic downstream processes such as increased cytosolic or extracellular DA release.

Excessive glutamate leads to large influxes of calcium that can trigger excitotoxicity leading to neurodegeneration and cell death (87). Chronic, mild calcium elevation can also dysregulate mitochondrial calcium buffering, and increase sensitivity to cytosolic calcium toxicity (39), and the route of calcium entry can also impact toxicity (88). Isradipine, a blocker of $Ca_v1.2$ and

$\text{Ca}_v1.3$ channels, failed to protect against hyperactivity-induced degeneration in our model (Figure S9). However, it is possible that other interventions against calcium-induced toxicity such as inhibition of NMDA receptors or the mitochondrial calcium uniporter may be protective. Alternatively, degeneration could be driven by the toxicity of excessive extracellular dopamine as a consequence of elevated activity. Future studies will explore these potential mechanisms.

If increased DA neuron activity contributes to degeneration in PD, then one might also expect gene expression changes in our model to mirror those seen in PD. Indeed, we found that mouse DEGs exhibited broadly congruent directional changes in the human dataset (Figure S10). This raises the possibility that similar mechanisms of degeneration and adaptation may be at play in both our mouse model and in PD. Interestingly, HSPA4 was significantly downregulated in the mouse and human datasets. Heat shock proteins broadly function to remove misfolded proteins, but the function of HSPA4 in neurons is not well described. However, in rat neural stem cells, HSPA4 is upregulated by selegiline, a type B monoamine oxidase inhibitor used to treat PD, and HSPA4 upregulation reduces ROS levels and mitochondrial DNA damage following hydrogen peroxide exposure (89 [↗](#), 90 [↗](#)). In MPTP-treated mice, deacetylation of HSPA4 is linked to decreased microglial activation and neuroinflammation (91 [↗](#)). In our activity model, decreased HSPA4 expression might indicate a failure to respond to inflammation, promoting neurodegeneration. Further studies will be required to investigate the role of HSPA4 and inflammation in chronic hyperactivation-induced toxicity.

Notably, we compared our mouse transcriptomic data at a stage when there is mild terminal loss and no somatic degeneration with early human PD patient tissues where there is limited dopamine neuron loss, raising confidence in the relevance of the comparison. However, it is important to consider that our Visium transcriptomic analysis has limited cell type resolution and our high resolution Xenium analysis is limited to nuclear transcripts. Therefore, important areas of future investigation include higher spatial resolution in the analysis of mouse tissue, and assessment of additional disease stages, brain areas, and cell types in human PD patient tissues.

In some cases, disease proteins or circuit changes may decrease neural activity. As such, it will be important to understand in which contexts activity increases or decreases, and also how decreasing DA activity influences degeneration. Indeed, if activity does promote degeneration, it will also be important to understand if decreasing DA neuron activity is protective, and if activity can be decreased or modulated without compromising motor function. Interestingly, chronic nicotine, which may mediate the protective association of smoking in PD (92 [↗](#), 93 [↗](#)), inhibits SNc DA neurons through agonism of nicotinic acetylcholine receptors expressed on presynaptic GABAergic terminals (94 [↗](#)). In addition, a proposed mechanism of action for the beneficial effects of deep brain stimulation (DBS) in PD is the inhibition of the subthalamic nucleus. As such, it will also be important to know how DBS influences DA neuron activity and if it has the potential to influence neurodegeneration. Interestingly, recent clinical trial data raise the possibility that DBS might slow disease progression when administered to early-stage patients (95 [↗](#)).

In summary, our data show that chronically increasing the activity of DA neurons can produce toxicity, and that SNc neurons may be more susceptible to this effect than VTA neurons. Considering that DA neuron activity may increase to compensate for other dying DA neurons, or in response to certain disease proteins, our data support the hypothesis that increased neural activity contributes to the pathophysiology of at least a subset of PD. Proving this, and determining how this evolves with disease stage, will be important goals for future research.

Methods

Experimental model and subject details

Mice

Mice were group-housed in a colony maintained with a standard 12-hour light/dark cycle and given food and water ad libitum. All mice received food on the cage floor. All animal experimental procedures were conducted in accordance with the Guide for the Care and Use of Laboratory Animals, as adopted by the National Institutes of Health, and with approval from the University of California, San Francisco Institutional Animal Care and Use Committee. All mice were housed in a state-of-the-art barrier facility managed by the UCSF Laboratory Animal Resource Center (LARC). Animal care and use in this research are covered under the UCSF “Assurance of Compliance with PHS Policy on Humane Care and Use of Laboratory Animals by Awardee Institutions” number A3400-01. Experiments were performed on age-matched mice. All mouse lines were maintained on a C57Bl/6 background (The Jackson Laboratory; RRID:IMSR_JAX:000664). DAT^{IREs}Cre mice (96%) were obtained from the Jackson laboratory.

Methods

Chemogenetics

A detailed protocol for mouse stereotaxic surgery can be found at <https://dx.doi.org/10.17504/protocols.io.b9kxr4xn>. DAT^{IREs}Cre homozygous mice (RRID:IMSR_JAX:006660) were injected bilaterally with 1 µL of rAAV8-hSyn-DIO-hM3Dq-mCherry (UNC Vector Core, AddGene, RRID: Addgene_50459) into the midbrain (−3.0 mm anterior-posterior, ±1.2 mm medial-lateral, −4.3 mm dorsal-ventral) using a stereotaxic frame (Kopf) and a microliter syringe (Hamilton).

A detailed protocol for monitoring mouse activity can be found at <https://dx.doi.org/10.17504/protocols.io.3bvl4qqe8vo5/v1>. Two weeks following surgery, animals were single-housed and habituated to wireless running wheels (MedAssociates ENV-047, RRID: SCR_024879) connected to a hub (MedAssociates DIG-807, RRID: SCR_024880,) for locomotion recordings and water bottles (Amazon) for drinking. A detailed protocol for validating responsiveness of DREADD-injected DAT^{IREs}Cre mice to CNO can be found at <https://dx.doi.org/10.17504/protocols.io.6qpvr3yrpvmk/v1>. Validation of responsiveness was done for all DREADD experiments, except for those unilaterally injected for chronic isradipine treatment (Figure S9). All mice had access to a running wheel during their treatment, except those for fiber photometry and chronic isradipine treatment. The mice used for photometry underwent 7 days of running wheel access approximately 3 weeks prior to the beginning of the experiment. The photometry headcaps sterically prevented mice from having access to a running wheel in their home cage. Mice used for chronic isradipine treatment were unilaterally injected with hM3Dq which does not lend to clear interpretation of running wheel data.

A detailed protocol for preparing and administering CNO via drinking water can be found at dx.doi.org/10.17504/protocols.io.n2bvj33oblk5/v1. CNO (NIMH, Tocris 4936) was administered ad libitum in 2% sucrose water at 300mg/L. CNO and vehicle waters were made fresh weekly and stored at 4°C protected from light.

Ex Vivo Recording

A detailed protocol for ex vivo electrophysiology can be found at dx.doi.org/10.17504/protocols.io.261gedqn7v47/v1. DAT^{IREs}Cre mice injected bilaterally with rAAV8-hSyn-DIO-hM3Dq-mCherry and chronically treated with vehicle or CNO for one week were provided to the electrophysiologist blind to in vivo treatment. Mice were deeply anesthetized with isoflurane, decapitated with a rodent guillotine, and the brains were removed. Horizontal brain

slices (150 μm thick) containing the SNc were prepared using a Vibratome (Campden Instruments, 7000smz-2). Slices were cut in ice cold aCSF solution containing (in mM): 119 NaCl, 2.5 KCl, 1.3 MgSO_4 , 1.0 NaH_2PO_4 , 2.5 CaCl_2 , 26.2 NaHCO_3 , and 11 glucose saturated with 95% O_2 –5% CO_2 and allowed to recover at 33°C in aCSF for at least 1 hr.

Slices were visualized under an *Axio Examiner A1* equipped with Dodt and IR optics, using a Zeiss Axiocam 506 mono and NeuroLucida 2023 (MBF Biosciences) software. Neurons were identified as DREADD-expressing prior to patching with fluorescent imaging of mCherry expressed concurrently with DREADDs. Whole-cell patch-clamp recordings were made at 33°C using 3–5 Mohm pipettes containing (in mM): 123 K-gluconate, 10 HEPES, 0.2 EGTA, 8 NaCl, 2 MgATP, and 0.3 Na_3GTP , pH 7.2, osmolarity adjusted to 275. Biocytin (0.1%) was also included in the internal solution to identify neurons after recordings where desired.

Recordings were made using Sutter IPA and SutterPatch v2.3.1 software (Sutter Instruments), filtered at 5 kHz and collected at 10 kHz. Liquid junction potentials were not corrected. I_h was recorded by voltage clamping cells at –60 mV and stepping to –40, –50, –70, –80, –90, –100, –110, and –120 mV. Cells were recorded in current-clamp mode ($I = 0$ pA) for measuring spontaneous firing rates and CNO responsivity. For CNO testing, spontaneous firing rate or resting membrane potential were monitored until a stable baseline was observed for at least 5 min. Then perfusion solution was switched 1 μM CNO for 5 min.

When recordings were completed, slices were drop fixed in 4% formaldehyde in PBS for at least 2 hr.

For quantifications, spontaneous firing rate was measured as the mean firing rate during the first 120 sec of whole cell recording. Once every 10 sec a brief hyperpolarizing pulse was applied, and the input resistance of the neuron was quantified from this test, averaged across the measurements made during the first 2 min of recording. AP waveform measurements were made from averages across at least 8 APs from this recording interval. I_h magnitude was quantified as the difference between the initial steady state response to the –120 mV step and the asymptote of the slow current sag.

Statistical analyses were completed in R (4.2.3), first testing whether data met the criteria for parametric statistical evaluation. Those datasets that met criteria were compared by unpaired Student's t-test. Those that did not meet these criteria were compared by permutation test. Code for the electrophysiology analysis can be found on GitHub at <https://github.com/eb-margolis-neuroscience-lab/R-general-Margolislab/blob/main/permtestsEBM.R> .

All salts and reagents were purchased from Sigma except CNO (Tocris).

Ex Vivo Data Analysis

Results are presented as mean $\pm$ S.E.M or with kernel density estimations. All but 1 neuron recorded in current clamp was quiescent during CNO testing. This neuron both depolarized and increased firing rate in response to CNO and therefore was included in the time course average figure. For all data parametric assumptions were tested to choose between t-test (parametric) or permutation (non-parametric) analysis.

Fiber photometry

A detailed protocol for collecting fiber photometry data can be found at [dx.doi.org](https://doi.org/10.17504/protocols.io.bp2l6xwxzlqe/v1) /10.17504/protocols.io.bp2l6xwxzlqe/v1. 3-month-old Ai148 mice (RRID:IMSR_JAX:030328) bred with DAT^{IRES} Cre were injected with rAAV8-hSyn-DIO-hM3Dq-mCherry as described above. After allowing 3 weeks for expression of the hM3Dq construct,

locomotion was tested using IP injection of CNO to determine robust hM3Dq expression. Mice that exhibited increased locomotion following CNO injection were then implanted with optical fibers (400 μ M, 0.48 NA). Mice were administered vehicle or CNO water for 14 days.

In vivo calcium data was acquired using a custom-built photometry system. An RZ5P fiber photometry processor (TDT, RRID: SCR_024878) and Synapse software (TDT, RRID: SCR_006307) were used to control LED output and acquire the photometry signal. Using this system, two LEDs were used to control GCaMP and isosbestic excitation (470 nm and 405 nm, respectively, Thorlabs). LEDs were sinusoidally modulated at 211 Hz (470 nm) and 531 Hz (405 nm) and entered a four-port fluorescence mini cube (Doric Lenses). The combined output was coupled to a fiber-optic patch cord (400 μ m, 0.48 NA, Thorlabs), which then mated to the fiberoptic cannula in the mouse brain. The emitted light was collected onto a visible femtowatt photoreceiver module (AC low, Newport) and sampled at 60 Hz. Photometry data was then extracted via proprietary TDT software using MATLAB (MathWorks, RRID: SCR_001622). Code for the fiber photometry analysis can be found on GitHub at <https://github.com/yoshitakasei/kreitzerPhotometry>.

Open Field Behavior

A detailed protocol for open field analysis can be found at [dx.doi.org/10.17504/protocols.io.36wgq33rklk5/v1](https://doi.org/10.17504/protocols.io.36wgq33rklk5/v1). Spontaneous locomotor activity was assessed while simultaneously recording fiber photometry data. Videos acquired during photometry sessions were analyzed using Ethovision software (Noldus, RRID: SCR_000441) to calculate total distance traveled and percent of time spent moving.

Chronic Isradipine Treatment and Plasma Detection

DAT^{ires}Cre mice were unilaterally inject in the right midbrain with AAV-hM3Dq as detailed above and in the left midbrain at the same coordinates with 1 μ L of sterile saline at 200nL per minute. Mice were allowed to heal for two weeks. Mice were then implanted with either placebo or isradipine slow-release pellets (Innovative Research of America). Pellets were designed to release 3 μ g/g/day of isradipine. Briefly, mice were anaesthetized with isoflurane and a placebo or isradipine pellet was implanted subcutaneously and the wound was closed with sutures. Administration of either vehicle or CNO water was initiated the same day as pellet implantation. Blood was drawn from anaesthetized mice and incubated in tubes containing K3 EDTA solution on ice. Tubes were centrifuged at 4°C at 10,000 *g* for 10 minutes. Collected supernatant was stored at –80°C until analyzed. Samples were sent to Charles River Laboratories for LC-MS/MS bioanalysis.

Monoamine Quantification

Striatal tissues flash frozen for Visium Spatial Transcriptomics (described below) were used for monoamine quantification. Briefly, flash frozen brains were embedded in OCT and mounted on a cryostat to collect dorsal striatal punches. Punches were stored at –80°C before shipment to the Vanderbilt Neurochemistry Core for catecholamine quantification by HPLC (27 [27](#)).

Immunohistochemistry

Animals were anesthetized with 2,2,2-tribromoethanol and perfused with PBS followed by 4% paraformaldehyde (PFA) in PBS. Intact brains were removed, post-fixed in 4% PFA overnight at 4°C and cryoprotected in 30% sucrose. 40- μ m-thick coronal sections were cut on a sliding microtome (Leica) and stored in cryoprotectant (30% Ethylene Glycol (Sigma), 30% Glycerol (Fisher scientific) in PBS).

A detailed protocol for immunofluorescence can be found at [dx.doi.org/10.17504/protocols.io.kxygx38owg8j/v1](https://doi.org/10.17504/protocols.io.kxygx38owg8j/v1). Brain sections were rinsed with PBS followed by 0.2% Triton X-100 in PBS. The sections were then transferred to a blocking solution containing 4% fetal bovine serum (JR Scientific) and 0.2% Triton X-100 in PBS for 1 hour. Following overnight

incubation in primary antibody, sections were rinsed in 0.2% Triton X-100 in PBS and incubated for 2 hours with a suitable secondary antibody. Sections were rinsed again in 0.2% Triton X-100 before mounting and coverslipping with antifade mounting medium (Vector Laboratories H1400, H1500).

A detailed protocol for peroxidase staining can be found at [dx.doi.org/10.17504/protocols.io.n92ldm127l5b/v1](https://doi.org/10.17504/protocols.io.n92ldm127l5b/v1). Sections were quenched with 3% H₂O₂ and 10% methanol in PBS and blocked in 10% fetal bovine calf serum, 3% BSA and 1% glycine in PBS with 0.5% Triton X-100. They were incubated with primary antibody followed by biotinylated secondary and subsequently streptavidin-conjugated horseradish peroxidase (HRP) (1:500; Vectastain ABC kit, Vector Laboratories). Immunostaining was visualized with hydrogen peroxide and 1 3,3'-diaminobenzidine (DAB, Sigma).

The following primary antibodies were used: rabbit anti-DsRed (1:1000; Takara, RRID: AB_3083500), rabbit anti-tyrosine hydroxylase (1:1000; AB152, Millipore, RRID: AB_390204), mouse anti-NeuN (Millipore MAB 377, RRID: AB_2298772), and chicken anti-mCherry (Abcam ab205402, RRID: AB_2722769). Secondary antibodies Alexafluor goat anti-mouse 647 (Thermo Fisher Scientific Cat# A-21235, RRID:AB_2535804), anti-rabbit 488 (Thermo Fisher Scientific Cat# A11034, RRID:AB_2576217), anti-rabbit 594 (Thermo Fisher Scientific Cat# A11037, RRID:AB_2534095), or anti-chicken 594 (Thermo Fisher Scientific Cat# A-11042, RRID:AB_2534099) IgG were used (1:500). A biotinylated goat anti-rabbit IgG (1:300, Vector Laboratories, BA-1000, RRID:AB_2313606) was used for peroxidase staining.

Stained samples were visualized using an automated fluorescence microscope (Keyence BZ7000) and a laser-scanning confocal microscope (Zeiss LSM880).

Stereology

Total numbers of TH- and mCherry-positive neurons were quantified blinded to groups. Region selection of SN and VTA was done under 5× magnification following the definition by Fu and colleagues (97). Imaging was done under 63× magnification by a Zeiss Imager microscope (Carl Zeiss Axio Imager A1) equipped with an XYZ computer-controlled motorized stage and an EOS rebel T5i Digital Camera (Canon), and counting was done using MBF Bioscience's Stereo Investigator (RRID: SCR_024705). Counting frame size was 60 × 60 μm and systematic random sampling (SRS) grid was 130 × 130 μm, with a section interval of 6.

Optical Density Analysis

A detailed protocol for optical density analysis can be found at [dx.doi.org/10.17504/protocols.io.81wgbxo2nlpk/v1](https://doi.org/10.17504/protocols.io.81wgbxo2nlpk/v1). Images of DAB-stained striatal sections were taken with an automated light microscope (Keyence) by a blinded experimenter. ImageJ (RRID: SCR_002285) was calibrated for optical density and subsequently used to draw ROIs around striatal areas and to measure mean optical density. The Allen Brain Atlas was used as a reference brain atlas (Allen Mouse Brain Atlas, mouse.brain-map.org and atlas.brain-map.org). Optical density values were background subtracted using an adjacent brain region with low TH expression levels, the piriform cortex.

Spatial Genomics

Mouse Visium Spatial Gene Expression

Spatial transcriptomics were acquired with Visium spatial gene expression kits (10X Genomics). Sample preparation, sample imaging, and library generation were completed in accordance with 10X Spatial Gene Expression protocols and as previously published (98). Briefly, fresh brain tissue was flash frozen in an isopentane bath cooled to -80°C using dry ice. The brain tissue was

then embedded in Optimal Cutting Temperature (OCT) compound (Tissue-Tek 62550-12). A cryostat was used to obtain a 10 μm thin section from the midbrain that was then mounted onto a 10X spatial gene expression slide. Sections were stained with TH, NeuN, and DsRed to visualize mCherry (striatal sections only), and Hoechst 33342 before imaging on a Leica Aperio Versa slide scanner. The cDNA libraries were generated at the Gladstone Genomics Core. Libraries were sequenced at the UCSF Center for Advanced Technology on an Illumina NovaSeq 600 on an SP flow cell. Alignment of the sequencing data with spatial data from the Visium slide was completed with the 10X Space Ranger software (10x Genomics, RRID: SCR_023571). The 10X Loupe Browser software (RRID: SCR_018555) was then used to identify six anatomical regions of interest: (1) SN, (2) VTA, and (3) thalamus in the midbrain, and (4) CP, (5) LSX, and (6) white matter tracts in the striatum. RNA capture areas corresponding to each anatomical region were selected for analysis based on their spatial proximity to the anatomical regions and on the expression of known genetic markers. SN and VTA genetic markers included TH, DAT, and VMAT2; thalamus markers included Prkcd, Ptpn3, and Synpo2; CP was identified with DARPP32; the LSX was identified with PRKCD; white matter was identified with MBP. Demarcation of SN and VTA was done according to Fu and colleagues (97). Capture areas that expressed high levels of astrocyte markers (Gfap, Aldh1l1) and microglia markers (Aif1, P2ry12) were excluded. SN and VTA capture areas also had to contain at least one complete DA neuron soma. Roughly 5-10 neuronal cell bodies fit into a single capture area for SN and VTA with 2-49 capture areas per VTA, 1-7 capture areas per SN, 0-39 capture areas per thalamus, 357-560 capture areas per CP, 6-93 capture areas per LSX, and 21-72 capture areas per white matter tracts. Gene expression levels were exported to GraphPad Prism and Microsoft Excel (Microsoft, RRID: SCR_016137). Gene rankings for hit analysis were established using the fold change score (FCS) and signal-to-noise score (SNS). Equations for these scores are given as:

$$SNS = \frac{\mu_{KO} - \mu_{WT}}{\sigma_{KO} + \sigma_{WT}} \times (-\log_{10} P)$$

$$FCS = \log_2 \left(\frac{\mu_{KO}}{\mu_{WT}} \right) \times (-\log_{10} P)$$

where μ is the average gene expression, σ is the standard deviation, and P is the p-value derived from a t-test. Genes with a p-value < 0.05 were highlighted as differentially expressed genes of interest. The expression levels for these genes of interest were probed in SN, VTA, and the thalamus to identify a subset of genes with DA region-specific changes. Pathway analysis was done on all hits that appeared in both scoring metrics using the Enrichr webtool (RRID: SCR_001575) (99–101). Code for Visium spatial transcriptomics analysis can be found on GitHub at <https://github.com/yoshitakasei/NakamuraVisiumR-Analysis>.

Principal Components Analysis

The R function `prcomp` was used for principal component analysis, with the median normalized gene expression level of each gene as the input. Only genes that were expressed in all regions (for SN and VTA) and whose expression was >0 were included.

Neuronal Activity Predictions

To quantify neuronal activity, we utilized NEUROeSTIMator (47), a deep learning model that generates a singular activity score following whole transcriptome reconstruction of 22 neuronal activity markers. This score has been shown to correlate significantly with electrophysiological features of hyper-excitability. We applied the NEUROeSTIMator algorithm to the Visium count matrix (species = “*mmusculus*”). Visium spatial transcriptomic capture areas within the SN and VTA were defined as described above, and neighboring spots were included using the `RegionNeighbors` (mode = “all_inner_outer”) function of `semmla` R statistical program (102). The predicted activity scores were normalized by division of the mean activity scores of the total tissue section they were derived. Group-wise normalized predicted activity scores were compared using the Kolmogorov-Smirnov test (KS-test).

Mouse Xenium Spatial Gene Expression

The 10X Genomics Xenium In Situ single cell spatial imaging platform was completed with Stanford Genomics in accordance with the 10X Xenium In Situ protocol (10X Genomics CG000581, CG000582, and CG000584 ([103](#))). Flash-frozen tissues embedded in OCT were mounted to Xenium Slides (10X Genomics 1000460) and imaged on the 10X Genomics Xenium Analyzer instrument (RRID:SCR_023910). The Xenium Analyzer performs Xenium Onboard Analysis (RRID:SCR_026158), which includes image pre-processing, puncta detection, transcript decoding, and quality score assignment. Transcript counts were restricted to the cell nucleus using the 10X Genomics Xenium Ranger v2.0.1 software suite.

A post-hoc immunofluorescence stain for mCherry was performed according to 10X Genomics protocols (10X Genomics CG000709) using the same DsRed primary and Alexa-594 secondary mentioned above. Images acquired at 40X using a laser-scanning confocal microscope (Zeiss LSM 880) were converted to pyramidal OME-TIFF files with QuPath v0.5.1 (RRID:SCR_018257). Images were aligned in the Xenium Explorer v3.2.0 (RRID:SCR_025847) according to 10X Genomics Image Alignment protocol adapted from Horn's absolute orientation algorithm ([104](#)).

Nuclei of dopaminergic neurons were identified via expression of TH, DAT, or VMAT2. Of the dopaminergic nuclei identified, only the nuclei surrounded by DsRed fluorescence (indicative of successful viral infection) were selected for transcript count analysis. Transcript counts for dopaminergic neurons infected with virus were extracted and exported using Seurat v5.1.0. The exported average transcript counts per nucleus were visualized in GraphPad Prism. Code for Xenium spatial transcriptomics analysis can be found on GitHub at https://github.com/yoshitakasei/Pilot_Xenium_w_mCherry_Marker.git.

Human Spatial Gene Expression

Cohort materials: Formalin-fixed paraffin-embedded (FFPE) midbrain sections were collected from individuals with pathologically verified early-stage Parkinson's Disease (ePD) (n=10) and individuals without any neurological or neuropathological conditions (n=10) through the New South Wales (NSW) Brain Banks, as detailed in Supplementary Table 2. The research protocol received ethical clearance from the University of Sydney Human Research Ethics Committee (approval number 2021/845). All PD subjects exhibited a positive response to levodopa and met the UK Brain Bank Clinical Criteria for PD diagnosis ([105](#)) without any other neurodegenerative disorders. Control subjects demonstrated no signs of Lewy body pathology, and ePD subjects had Braak stage IV Lewy body pathology, in accordance with established criteria ([106](#), [107](#)).

Sequencing and data processing: FFPE sections were stained using TH antibody (BioLegend, cat# 818008, 1:50 RRID: AB_2801155) and processed using the Nanostring GeoMx® Digital Spatial Profiler (RRID: SCR_021660), as per manufacturer instructions, to obtain TH+ masked transcripts. Sequencing libraries for the whole transcriptome were constructed using the Human Whole Transcriptome Atlas (GeoMx Hu WTA) following manufacturer instructions. Technical replicates were consolidated using the Linux 'cat' command, and alignment and feature counting were performed using the GeoMx NGS analysis pipeline (version 2.0.21) executed on the Illumina BaseSpace platform (Illumina, RRID: SCR_011881). Quality control was implemented using R statistical programming language with the cutoffs; minSegmentReads: 1000, percentTrimmed: 80%, percentStitched: 80%, percentAligned: 75%, percentSaturation: 50%, minNegativeCount: 1, maxNTCCount: 10,000, minNuclei: 20, minArea: 1000. The minimum gene detection rate across all samples was set at 1%. The minimum gene detection rate per sample was set to 1%. Recently, Van Hifft and colleagues reviewed the recommended Nanostring GeoMx Q3 normalization technique ([108](#)), observing that Q3 normalization failed to correct for large differences in the signal (gene expression) to noise (neg probes) observed between samples and recommended quantile normalization. We performed preliminary experiments of 5 normalization techniques; Q3

normalization, background normalization, quantile normalization, SCTransform and normalizing for total library size. We observed that quantile normalization displayed the lowest absolute MA plot correlation and least significance following Kolmogorov Smirnov test. Hence, quantile normalization was implemented, and batches (slides) were corrected using Harmony (109), DEGs identified by spatial transcriptomic analysis of mouse model were evaluated between control and ePD TH+ masked Regions of Interest (ROIs) from the SNc ventral tier. The congruence between directional changes of mouse and human was quantified using a chi-square test. To assess differential gene expression between control (CTR) and Parkinson's disease (PD) samples, we employed the 'limma voom' methodology, incorporating covariates diagnosis, age, post-mortem delay (PMD), RNA integrity number equivalent (DV200), sex, DSP processing plate, and brain bank ID (110). This analysis was conducted using the R statistical software environment with all scripts are available at https://github.com/zchatt/ASAP-SpatialTranscriptomics/tree/Rademacher_2024 and processed and raw datasets are available at [10.5281/zenodo.10499186](https://zenodo.org/record/10499186).

Quantification and Statistical Analysis

All statistical analyses including the n, what n represents, description of error bars, statistical tests used and level of significance are found in the figure legends. Two-way repeated-measures ANOVA followed by Holm-Sidak multiple comparisons was used for comparing vehicle vs CNO groups. One-way ANOVA followed by Holm-Sidak multiple comparisons was used for comparing multiple groups in mouse spatial transcriptomic data. Three-way ANOVA followed by Holm-Sidak multiple comparisons was used for assessing chronic isradipine histology. T-test or permutation (non-parametric) analysis was used for ex vivo electrophysiology data. Mouse differentially expressed genes were ranked according to the defined SNS and FCS metrics. Congruence between directional changes of mouse and human was quantified using a chi-square test. 'Limma voom' methodology was used to assess differential gene expression between control and PD samples, incorporating covariates (see Human Spatial Gen Expression). All analyses were performed using GraphPad Prism version 9 (RRID: SCR_002798), Microsoft Excel, and R version 4.2.2 (RRID: SCR_000432).

Data, Materials, and Software Availability

Datasets are available at <https://doi.org/10.5281/zenodo.10499027>. All other data are included in the manuscript and/or supporting information.

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Additional files

Supplementary Figures and Tables [↗](#)

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Reviewer #1 (Public review):

Summary:

In this manuscript, the authors investigated the effect of chronic activation of dopamine neurons using chemogenetics. Using Gq-DREADDs, the authors chronically activated midbrain dopamine neurons and observed that these neurons, particularly their axons, exhibit increased vulnerability and degeneration, resembling the pathological symptoms of Parkinson's disease. Baseline calcium levels in midbrain dopamine neurons were also significantly elevated following the chronic activation. Lastly, to identify cellular and circuit-level changes in response to dopaminergic neuronal degeneration caused by chronic activation, the authors employed spatial genomics (Visium) and revealed comprehensive changes in gene expression in the mouse model subjected to chronic activation. In conclusion, this study presents novel data on the consequences of chronic hyperactivation of midbrain dopamine neurons.

Strengths:

This study provides direct evidence that the chronic activation of dopamine neurons is toxic and gives rise to neurodegeneration. In addition, the authors achieved the chronic activation of dopamine neurons using water application of clozapine-N-oxide (CNO), a method not commonly employed by researchers. This approach may offer new insights into pathophysiological alterations of dopamine neurons in Parkinson's disease. The authors also utilized state-of-the-art spatial gene expression analysis, which can provide valuable information for other researchers studying dopamine neurons. They also presented a substantial number of intriguing ideas in their discussion, which are worth further investigation.

Weaknesses:

Although not fully supported by data, the authors provided a well-explained rationale and proposed possible mechanisms for dopamine neuron degeneration due to chronic activation

in their results and discussion.

Comments on revised version:

The authors have adequately addressed most of my comments, and I have no further concerns.

<https://doi.org/10.7554/eLife.98775.2.sa3>

Reviewer #2 (Public review):

Rademacher et al. present a paper showing that chronic chemogenetic excitation of dopaminergic neurons in the mouse midbrain results in differential degeneration of axons and somas across distinct regions (SNc vs VTA). These findings are important for two reasons: 1. This approach can be used as a mouse model for Parkinson's Disease without the need for the infusion of toxins (e.g. 6-OHDA or MPTP). This mouse model also has the advantage of showing a axon-first degeneration over an experimentally-useful time course (2-4 weeks). 2. The findings that direct excitation of dopaminergic neurons causes differential degeneration sheds light on the mechanisms of dopaminergic neuron selective vulnerability. The evidence that activation of dopaminergic neurons causes degeneration, alters motor behavior, and alters mRNA expression is convincing. This is an exciting and important paper and will have an impact on the Parkinson's Disease field.

Strengths:

This is an exciting and important paper and will have an impact on the Parkinson's Disease field.

It presents a new highly useful mouse model of PD.

The paper compares mouse transcriptomics with human patient data.

It shows that selective degeneration can occur across the midbrain dopaminergic neurons even in the absence of a genetic, prion, or toxin neurodegeneration mechanism.

Weaknesses:

The authors have addressed all my concerns. This is an interesting, important, and carefully-controlled study.

<https://doi.org/10.7554/eLife.98775.2.sa2>

Reviewer #3 (Public review):

Summary:

In this manuscript, Rademacher and colleagues examined the effect on the integrity of the dopamine system in mice of chronically stimulating dopamine neurons using a chemogenetic approach. They find that one to two weeks of constant exposure to the chemogenetic activator CNO leads to a decrease in the density of tyrosine hydroxylase staining in striatal brain sections and to a small reduction of the global population of tyrosine hydroxylase positive neurons in the ventral midbrain. They also report alterations in gene expression in both regions using a spatial transcriptomics approach. Globally, the work is well done and valuable and some of the conclusions are interesting. However, the conceptual advance is perhaps a bit limited in the sense that there is extensive previous work in the literature showing that excessive depolarization of multiple types of neurons associated with

intracellular calcium elevations promotes neuronal degeneration. The present work adds to this by showing evidence of a similar phenomenon in dopamine neurons. In terms of the mechanisms explaining the neuronal loss observed after 2 to 4 weeks of chemogenetic activation, it would be important to consider that dopamine neurons are known from a lot of previous literature to undergo a decrease in firing through a depolarization-block mechanism when chronically depolarized. Is it possible that such a phenomenon explains much of the results observed in the present study? It would be important to consider this in the manuscript. The relevance to Parkinson's disease (PD) is also not totally clear because there is not a lot of previous solid evidence showing that the firing of dopamine neurons is increased in PD, either in human subjects or in mouse models of the disease.

Comments on revisions:

The authors have done a good job at revising the manuscript. The revised manuscript better frames the results in the context of previous literature.

<https://doi.org/10.7554/eLife.98775.2.sa1>

Author response:

The following is the authors' response to the original reviews

Reviewer #1 (Public Review):

Summary:

In this manuscript, the authors investigated the effect of chronic activation of dopamine neurons using chemogenetics. Using Gq-DREADDs, the authors chronically activated midbrain dopamine neurons and observed that these neurons, particularly their axons, exhibit increased vulnerability and degeneration, resembling the pathological symptoms of Parkinson's disease. Baseline calcium levels in midbrain dopamine neurons were also significantly elevated following the chronic activation. Lastly, to identify cellular and circuit-level changes in response to dopaminergic neuronal degeneration caused by chronic activation, the authors employed spatial genomics (Visium) and revealed comprehensive changes in gene expression in the mouse model subjected to chronic activation. In conclusion, this study presents novel data on the consequences of chronic hyperactivation of midbrain dopamine neurons.

Strengths:

This study provides direct evidence that the chronic activation of dopamine neurons is toxic and gives rise to neurodegeneration. In addition, the authors achieved the chronic activation of dopamine neurons using water application of clozapine-N-oxide (CNO), a method not commonly employed by researchers. This approach may offer new insights into pathophysiological alterations of dopamine neurons in Parkinson's disease. The authors also utilized state-of-the-art spatial gene expression analysis, which can provide valuable information for other researchers studying dopamine neurons. Although the authors did not elucidate the mechanisms underlying dopaminergic neuronal and axonal death, they presented a substantial number of intriguing ideas in their discussion, which are worth further investigation.

We thank the reviewer for these positive comments.

Weaknesses:

Many claims raised in this paper are only partially supported by the experimental results. So, additional data are necessary to strengthen the claims. The effects of chronic activation of dopamine neurons are intriguing; however, this paper does not go beyond reporting phenomena. It lacks a comprehensive explanation for the degeneration of dopamine neurons and their axons. While the authors proposed possible mechanisms for the degeneration in their discussion, such as differentially expressed genes, these remain experimentally unexplored.

We thank the reviewer for this review. We do believe that the manuscript has a substantial mechanistic component, as the central experiments involve direct manipulation of neuronal activity, and we show an increase in calcium levels and gene expression changes in dopamine neurons that coincide with the degeneration. However, we agree that deeper mechanistic investigation would strengthen the conclusions of the paper. We have executed several important revisions, including the addition of CNO behavioral controls, manipulation of intracellular calcium using isradipine, additional transcriptomics experiments and further validation of findings. We believe that these additions significantly bolster the conclusions of the paper.

Reviewer #2 (Public Review):

Summary:

Rademacher et al. present a paper showing that chronic chemogenetic excitation of dopaminergic neurons in the mouse midbrain results in differential degeneration of axons and somas across distinct regions (SNc vs VTA). These findings are important. This mouse model also has the advantage of showing an axon-first degeneration over an experimentally-useful time course (2-4 weeks). 2. The findings that direct excitation of dopaminergic neurons causes differential degeneration sheds light on the mechanisms of dopaminergic neuron selective vulnerability. The evidence that activation of dopaminergic neurons causes degeneration and alters mRNA expression is convincing, as the authors use both vehicle and CNO control groups, but the evidence that chronic dopaminergic activation alters circadian rhythm and motor behavior is incomplete as the authors did not run a CNO-control condition in these experiments.

Strengths:

This is an exciting and important paper.

The paper compares mouse transcriptomics with human patient data.

It shows that selective degeneration can occur across the midbrain dopaminergic neurons even in the absence of a genetic, prion, or toxin neurodegeneration mechanism.

We thank the reviewer for these comments.

Weaknesses:

Major concerns:

(1) The lack of a CNO-positive, DREADD-negative control group in the behavioral experiments is the main limitation in interpreting the behavioral data. Without knowing whether CNO on its own has an impact on circadian rhythm or motor activity, the certainty that dopaminergic hyperactivity is causing these effects is lacking.

We thank the reviewer for this important recommendation. Although the initial version showed that CNO does not produce degeneration of DA neuron terminals, it did not exclude a

contribution to the behavioral changes. To address this, we now include a cohort of DREADD free non-injected mice treated with either vehicle or CNO (Figure S1C). We found that on its own, CNO did not significantly impact either light cycle or dark cycle running. Together these results along with the lack of degeneration observed with CNO treatment in non-DREADD mice (Figure 2D) support that our behavioral and histological results are the result of dopamine neuron activation.

(2) One of the most exciting things about this paper is that the SNc degenerates more strongly than the VTA when both regions are, in theory, excited to the same extent. However, it is not perfectly clear that both regions respond to CNO to the same extent. The electrophysiological data showing CNO responsiveness is only conducted in the SNc. If the VTA response is significantly reduced vs the SNc response, then the selectivity of the SNc degeneration could just be because the SNc was more hyperactive than the VTA. Electrophysiology experiments comparing the VTA and SNc response to CNO could support the idea that the SNc has substantial intrinsic vulnerability factors compared to the VTA.

We agree that additional electrophysiology conducted in the VTA dopamine neurons would meaningfully add to our understanding of the selective vulnerability in this model, and have completed these experiments in the revision (Figure 1, Figure S2). We now show that *in vivo* treatment with CNO causes some of the same physiological changes in VTA dopamine neurons as we found in SNc dopamine neurons, including an increased spontaneous firing rate, and a similar decrease in responsiveness to CNO in the slice recordings. Together these observations support the conclusion that SNc axons are intrinsically more vulnerable to increased activity than VTA dopamine axons.

(3) The mice have access to a running wheel for the circadian rhythm experiments. Running has been shown to alter the dopaminergic system (Bastoli et al., 2022) and so the authors should clarify whether the histology, electrophysiology, fiber photometry, and transcriptomics data are conducted on mice that have been running or sedentary.

We have clarified which mice had access to a running wheel in the methods of our revision. Briefly, mice for histology, electrophysiology, and transcriptomics all had access to a running wheel during their treatment. The mice used for photometry underwent about 7 days of running wheel access approximately 3 weeks prior to the beginning of the experiment. The photometry headcaps prevented mice from having access to a running wheel in their home cage. Mice used for non-responder and non-hM3Dq (CNO alone) experiments also had access to a running wheel during their treatment. Mice used for the isradipine experiment did not have access to a running wheel, as the number of mice was too large and while unilateral hM3Dq expression allows for within-animal controls, it does not lend to clear interpretation of running wheel data.

Reviewer #3 (Public Review):

Summary:

In this manuscript, Rademacher and colleagues examined the effect on the integrity of the dopamine system in mice of chronically stimulating dopamine neurons using a chemogenetic approach. They find that one to two weeks of constant exposure to the chemogenetic activator CNO leads to a decrease in the density of tyrosine hydroxylase staining in striatal brain sections and to a small reduction of the global population of tyrosine hydroxylase positive neurons in the ventral midbrain. They also report alterations in gene expression in both regions using a spatial transcriptomics approach. Globally, the work is well done and valuable and some of the conclusions are interesting. However, the conceptual advance is perhaps a bit limited in the sense that there is

extensive previous work in the literature showing that excessive depolarization of multiple types of neurons associated with intracellular calcium elevations promotes neuronal degeneration. The present work adds to this by showing evidence of a similar phenomenon in dopamine neurons.

We thank the reviewer for the careful and thoughtful review of our manuscript.

While extensive depolarization and associated intracellular calcium elevations promote degeneration generally, we emphasize that the process we describe is novel. Indeed, prior studies delivering chronic DREADDs to vulnerable neurons in models of Alzheimer's disease did not detect an increase in neurodegeneration, despite seeing changes in protein aggregation (e.g. Yuan and Grutzendler, J Neurosci 2016, PMID: 26758850; Hussaini et al., PLOS Bio 2020, PMID: 32822389). Further, a critical finding from our study is that in our paradigm, this stressor does not impact all dopamine neurons equally, as the SNc DA neurons are more vulnerable than VTA DA neurons, mirroring selective vulnerability characteristic of Parkinson's disease. This is consistent with a large body of literature that SNc dopamine neurons are less capable of handling large energetic and calcium loads compared to neighboring VTA neurons, and the finding that chronically altered activity is sufficient to drive this preferential loss is novel. In addition, we are not aware of prior studies that have chronically activated DREADDs over several weeks to produce neurodegeneration.

In terms of the mechanisms explaining the neuronal loss observed after 2 to 4 weeks of chemogenetic activation, it would be important to consider that dopamine neurons are known from a lot of previous literature to undergo a decrease in firing through a depolarization-block mechanism when chronically depolarized. Is it possible that such a phenomenon explains much of the results observed in the present study? It would be important to consider this in the manuscript.

Thank you for this comment. As discussed in greater detail in the “comments on results section” below, our data suggests this isn't a prominent feature in our model. However, we cannot rule out a contribution of depolarization block, and have expanded on the discussion of this possibility in the revised manuscript.

The relevance to Parkinson's disease (PD) is also not totally clear because there is not a lot of previous solid evidence showing that the firing of dopamine neurons is increased in PD, either in human subjects or in mouse models of the disease. As such, it is not clear if the present work is really modelling something that could happen in PD in humans.

We completely agree that evidence of increased dopamine neuron activity from human PD patients is lacking, and the little data that exists is difficult to interpret without human controls. However, as we outline in the manuscript, multiple lines of evidence suggest that the activity level of dopamine neurons almost certainly does change in PD. Therefore, it is very important that we understand how changes in the level of neural activity influence the degeneration of DA neurons. In this paper we examine the impact of increased activity. Increased activity may be compensatory after initial dopamine neuron loss, or may be an initial driver of death (Rademacher & Nakamura, Exp Neurol 2024, PMID: 38092187). In addition to the human and rodent data already discussed in the manuscript, additional support for increased activity in PD models include:

- Elevated firing rates in asymptomatic MitoPark mice (Good et al., FASEB J 2011, PMID: 21233488)
- Increased frequency of spontaneous firing in patient-derived iPSC dopamine neurons and primary mouse dopamine neurons that overexpress synuclein (Lin et al., Acta Neuropath Comm 2021, PMID: 34099060)

- Increased spontaneous firing in dopamine neurons of rats injected with synuclein preformed fibrils compared to sham (Tozzi et al., Brain 2021, PMID: 34297092)

We have included citation of these important examples in our revision. In our model, we have found that chronic hyperactivity causes a substantial loss of nigral DA terminals while mesolimbic terminals are relatively spared (Figure 2), and that striatal DA levels are markedly decreased (Figure S6), phenomena that are hallmarks of Parkinson's disease.

There are additional levels of complexity to accurately model changes in PD, which may differ between subtypes of the disease, the disease stage, and the subtype of dopamine neuron. Our study models a form of increased intrinsic activity, and interpretation of our results will be facilitated as we learn more about how the activity of DA neurons changes in humans in PD. Similarly, in future studies, it will also be important to study the impact of decreasing DA neuron activity.

Comments on the introduction:

The introduction cites a 1990 paper from the lab of Anthony Grace as support of the fact that DA neurons increase their firing rate in PD models. However, in this 1990 paper, the authors stated that: "With respect to DA cell activity, depletions of up to 96% of striatal DA did not result in substantial alterations in the proportion of DA neurons active, their mean firing rate, or their firing pattern. Increases in these parameters only occurred when striatal DA depletions exceeded 96%." Such results argue that an increase in firing rate is most likely to be a consequence of the almost complete loss of dopamine neurons rather than an initial driver of neuronal loss. The present introduction would thus benefit from being revised to clarify the overriding hypothesis and rationale in relation to PD and better represent the findings of the paper by Hollerman and Grace.

We agree that the findings of Hollerman and Grace support compensatory changes in dopamine neuron activity in response to loss of dopamine neurons, rather than informing whether dopamine neuron loss can also be an initial driver of activity. Importantly, while significant changes to burst firing were not seen until almost complete loss of dopamine neurons, these recordings were made in anesthetized rats which may not be representative of neural activity in awake animals. We adjusted the text so that this is no longer referred to as 'partial' loss. At the same time, we point out that the results of other studies on this point are mixed: a 50% reduction in dopamine neurons didn't alter firing rate or bursting (Harden and Grace, J Neurosci 1995, PMID: 7666198; Bilbao et al., Brain Res 2006, PMID: 16574080), while a 40% loss was found to increase firing rate and bursting (Chen et al., Brain Res 2009, PMID: 19545547) and larger reductions alter burst firing (Hollerman & Grace, Brain Res 1990, PMID: 2126975; Stachowiak et al., J Neurosci 1987, PMID: 3110381). Importantly, even if compensatory, such late-stage increases in dopamine neuron activity may contribute to disease progression and drive a vicious cycle of degeneration in surviving neurons. In addition, we also don't know how the threshold of dopamine neuron loss and altered activity may differ between mice and humans, and PD patients do not present with clinical symptoms until ~30-60% of nigral neurons are lost (Burke & O'Malley, Exp Neurol 2013, PMID: 22285449; Shulman et al., Annu Rev Pathol 2011, PMID: 21034221).

Other lines of evidence support the potential role of hyperactivity in disease initiation, including increased activity before dopamine neuron loss in MitoPark mice (Good et al., FASEB J 2011, PMID: 21233488), increased spontaneous firing in patient-derived iPSC dopamine neurons (Lin et al., Acta Neuropath Comm 2021, PMID: 34099060), and increased activity observed in genetic models of PD (Bishop et al., J Neurophysiol 2010, PMID: 20926611; Regoni et al., Cell Death Dis 2020, PMID: 33173027).

It would be good that the introduction refers to some of the literature on the links between excessive neuronal activity, calcium, and neurodegeneration. There is a large literature on this and referring to it would help frame the work and its novelty in a broader context.

We agree that a discussion of hyperactivity, calcium, and neurodegeneration would benefit the introduction. Accordingly, we have expanded on our citation of this literature in both the introduction and discussion sections. However, we believe that the novelty of our study lies in: 1) a chronic chemogenetic activation paradigm via drinking water, 2) demonstrating selective vulnerability of dopamine neurons as a result of altering their activity/excitability alone, and 3) comparing mouse and human spatial transcriptomics.

Comments on the results section:

The running wheel results of Figure 1 suggest that the CNO treatment caused a brief increase in running on the first day after which there was a strong decrease during the subsequent days in the active phase. This observation is also in line with the appearance of a depolarization block.

The authors examined many basic electrophysiological parameters of recorded dopamine neurons in acute brain slices. However, it is surprising that they did not report the resting membrane potential, or the input resistance. It would be important that this be added because these two parameters provide key information on the basal excitability of the recorded neurons. They would also allow us to obtain insight into the possibility that the neurons are chronically depolarized and thus in depolarization block.

We do report the input resistance in Figure S1C (now Figure S2A, S2B), which was unchanged in CNO-treated animals compared to controls. We did not previously report the resting membrane potential because many of the DA neurons were spontaneously firing. In the revision, we now report the initial membrane potential on first breaking into the cell for the whole cell recordings, which did not vary between groups (Figure S2). This is still influenced by action potential activity, but is the timepoint in the recording least impacted by dialyzing the neuron with the internal solution, which might alter the intracellular concentrations of ions. We observed increased spontaneous action potential activity *ex vivo* in slices from CNO-treated mice (Figure 1D), thus at least under these conditions these dopamine neurons are not in depolarization block. We also did not see strong evidence of changes in other intrinsic properties of the neurons with whole cell recordings (e.g. Figure S2). Overall, our electrophysiology experiments are not consistent with the depolarization block model, at least not due to changes in the intrinsic properties of the neurons. Although our *ex vivo* findings cannot exclude a contribution of depolarization block *in vivo*, we do show that CNO-treated mice removed from their cages for open field testing continue to have a strong trend for increased activity for approximately 10 days (Figure S4B). This finding is also consistent with increased activity of the DA neurons. We have added discussion of these important considerations in the revision.

It is great that the authors quantified not only TH levels but also the levels of mCherry, coexpressed with the chemogenetic receptor. This could in principle help to distinguish between TH downregulation and true loss of dopamine neuron cell bodies. However, the approach used here has a major caveat in that the number of mCherry-positive dopamine neurons depends on the proportion of dopamine neurons that were infected and expressed the DREADD and this could very well vary between different mice. It is very unlikely that the virus injection allowed to infect 100% of the neurons in the VTA and SNC. This could for example explain in part the mismatch between the number of VTA dopamine neurons counted in panel 2G when comparing TH and mCherry counts. Also, I

see that the mCherry counts were not provided at the 2-week time point. If the mCherry had been expressed genetically by crossing the DAT-Cre mice with a floxed fluorescent reporter mice, the interpretation would have been simpler. In this context, I am not convinced of the benefit of the mCherry quantifications. The authors should consider either removing these results from the final manuscript or discussing this important limitation.

We thank the reviewer for this comment, and we agree that this is a caveat of our mCherry quantification. Quantitation of the number of mCherry+ DA neurons specifically informs the impact on transduced DA neurons, and mCherry appears to be less susceptible to downregulation versus TH. As the reviewer points out, it carries the caveat that there is some variability between injections. Our control animals give us an indicator of injection variability, which is likely substantial and prevents us from detecting more subtle changes. Nonetheless, we believe that it conveys useful complementary data. We discuss this caveat in our revision. Note that mCherry was not quantified at the two-week timepoint because there is no loss of TH+ cells at that time.

Although the authors conclude that there is a global decrease in the number of dopamine neurons after 4 weeks of CNO treatment, the post-hoc tests failed to confirm that the decrease in dopamine number was significant in the SNc, the region most relevant to Parkinson's. This could be due to the fact that only a small number of mice were tested. A "n" of just 4 or 5 mice is very small for a stereological counting experiment. As such, this experiment was clearly underpowered at the statistical level. Also, the choice of the image used to illustrate this in panel 2G should be reconsidered: the image suggests that a very large loss of dopamine

neurons occurred in the SNc and this is not what the numbers show. A more representative image should be used.

We agree that the stereology experiments were performed on relatively small numbers of animals, such that only robust effects would be detected. Combined with the small effect size, this may have contributed to the post-hoc tests showing a trend of $p=0.1$ for both the TH and mCherry dopamine cell counts in the SN at 4 weeks. Given this small effect size, we would indeed need much larger groups to better discern these changes. Stereology is an intensive technique, and we have therefore elected to focus on terminal loss. We have also replaced panel 2G with a more representative CNO image.

In Figure 3, the authors attempt to compare intracellular calcium levels in dopamine neurons using GCaMP6 fluorescence. Because this calcium indicator is not quantitative (unlike ratiometric sensors such as Fura2), it is usually used to quantify relative changes in intracellular calcium. The present use of this probe to compare absolute values is unusual and the validity of this approach is unclear. This limitation needs to be discussed. The authors also need to refer in the text to the difference between panels D and E of this figure. It is surprising that the fluctuations in calcium levels were not quantified. I guess the hypothesis was that there should be more or larger fluctuations in the mice treated with CNO if the CNO treatment led to increased firing. This needs to be clarified.

We thank the reviewer for this comment. We understand that this method of comparing absolute values is unconventional. However, these animals were tested concurrently on the same system, and a clear effect on the absolute baseline was observed. We have included a caveat of this in our discussion. Panel D of this figure shows the raw, uncorrected photometry traces, whereas panel E shows the isosbestic corrected traces for the same recording. In panel E, the traces follow time in ascending order. We have also included frequency and amplitude

data for these recordings (Figure S4A), along with discussion of the significance of these findings.

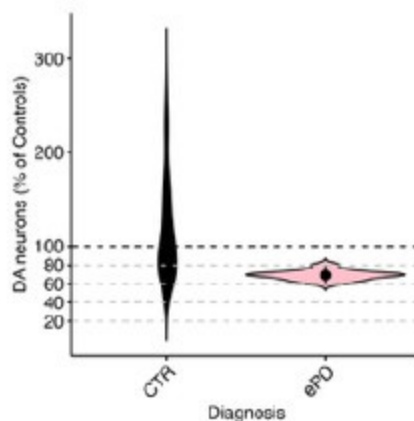
*Although the spatial transcriptomic results are intriguing and certainly a great way to start thinking about how the CNO treatment could lead to the loss of dopamine neurons, the presented results, the focusing of some broad classes of differentially expressed genes and on some specific examples, do not really suggest any clear mechanism of neurodegeneration. It would perhaps be useful for the authors to use the obtained data to validate that a state of chronic depolarization was indeed induced by the chronic CNO treatment. Were genes classically linked to increased activity like *cfos* or *bdnf* elevated in the SNc or VTA dopamine neurons? In the striatum, the authors report that the levels of *DARPP32*, a gene whose levels are linked to dopamine levels, are unchanged. Does this mean that there were no major changes in dopamine levels in the striatum of these mice?*

While levels of *DARPP32* mRNA were unchanged, our additional HPLC data show strong decreases in striatal dopamine in hyperactivated mice. We do not see strong changes in classic activity-related genes (data not shown), however these genes may behave differently in the context of chronic hyperactivity and ongoing degeneration. Instead, we employed NEUROeSTIMator (Bahl et al., Nature Comm. 2024, PMID: 38278804), a deep learning method to predict neural activation based on transcriptomic data. We found that predicted activity scores were significantly higher in GqCNO dopaminergic regions compared to controls (Figure X). Indeed, some of the genes used within the model to predict activity are immediate early genes eg. *c-fos*.

The usefulness of comparing the transcriptome of human PD SNc or VTA sections to that of the present mouse model should be better explained. In the human tissues, the transcriptome reflects the state of the tissue many years after extensive loss of dopamine neurons. It is expected that there will be few if any SNc neurons left in such sections. In comparison, the mice after 7 days of CNO treatment do not appear to have lost any dopamine neurons. As such, how can the two extremely different conditions be reasonably compared? Our mouse model and human PD progress over distinct timescales, as is the case with essentially all mouse models of neurodegenerative diseases. Nonetheless, in our view there is still great value in comparing gene expression changes in mouse models with those in human disease. It seems very likely that the same pathologic processes that drive degeneration early in the disease continue to drive degeneration later in the disease. Note that we have tried to address the discrepancy in time scales in part by comparing our mouse model to early PD samples when there is more limited SNc DA neuron loss (see the proportion of DA neurons within the areas of human tissues we selected for sampling in Author response image 1). Therefore, we can indeed use spatial transcriptomics to compare dopamine neurons from mice with initial degeneration to those in patients where degeneration is ongoing.

Author response image 1.

Violin plot of DA neuron proportions sampled within the vulnerable SNV (deconvoluted RCTD method used in unmasked tissue sections of the SNV). Control and early PD subjects.



Comments on the discussion:

In the discussion, the authors state that their calcium photometry results support a central role of calcium in activity-induced neurodegeneration. This conclusion, although plausible because of the very broad pre-existing literature linking calcium elevation (such as in excitotoxicity) to neuronal loss, should be toned down a bit as no causal relationship was established in the experiments that were carried out in the present study.

Our model utilizes hM3Dq-DREADDs that function by activating Gq pathways that are classically expected to increase intracellular calcium to increase neuronal excitability. Indeed in slices from mice that were not treated with CNO, acute CNO application caused depolarizations (Figure 1E) that can be due to an increase in intracellular calcium and also cause increases in intracellular calcium. Additionally, our results show increased calcium by fiber photometry and changes to calcium-related genes, suggesting a causal relation and crucial role of calcium in the mechanism of degeneration. However, we agree that we have not experimentally proven this point. Indeed, a small preliminary experiment with chronic isradipine failed to show protection, although it lacked power to detect a partial effect. We have acknowledged this in the text, and also briefly consider other mechanisms such as increased dopamine levels that could also mediate the toxicity.

In the discussion, the authors discuss some of the parallel changes in gene expression detected in the mouse model and in the human tissues. Because few if any dopamine neurons are expected to remain in the SNc of the human tissues used, this sort of comparison has important conceptual limitations and these need to be clearly addressed.

As discussed, we sampled SN DA neurons in early PD (see Author response image 1), and in our view there is great value for such comparisons.

A major limitation of the present discussion is that it does not discuss the possibility that the observed phenotypes are caused by the induction of a chronic state of depolarization block by the chronic CNO treatment. I encourage the authors to consider and discuss this hypothesis.

As discussed above, our analyses of DA neuron firing in slices and open field testing to date do not support a prominent contribution of depolarization block with chronic CNO treatment. However, we cannot rule out this hypothesis, therefore we have included

additional electrophysiology experiments and have added discussion of this important consideration.

Also, the authors need to discuss the fact that previous work was only able to detect an increase in the firing rate of dopamine neurons after more than 95% loss of dopamine neurons. As such, the authors need to clearly discuss the relevance of the present model to PD. Are changes in firing rate a driver of neuronal loss in PD, as the authors try to make the case here, or are such changes only a secondary consequence of extensive neuronal loss (for example because a major loss of dopamine would lead to reduced D2 autoreceptor activation in the remaining neurons, and to reduced autoreceptor-mediated negative feedback on firing). This needs to be discussed.

As discussed above, while increases in dopamine neuron activity may be compensatory after loss of neurons, the precise percentage required to induce such compensatory changes is not defined in mice and varies between paradigms, and the threshold level is not known in humans. We also reiterate that a compensatory increase in activity could still promote the degeneration of critical surviving DA neurons, whose loss underlies the substantial decline in motor function that typically occurs over the course of PD. Moreover, there are also multiple lines of evidence to suggest that changes in activity can initiate and drive dopamine neuron degeneration (Rademacher & Nakamura, Exp Neurol 2024). For example, overexpression of synuclein can increase firing in cultured dopamine neurons (Dagra et al., NPJ Parkinsons Dis 2021, PMID: 34408150), while mice expressing mutant Parkin have higher mean firing rates (Regoni et al., Cell Death Dis 2020, PMID: 33173027). Similarly, an increased firing rate has been reported in the MitoPark mouse model of PD at a time preceding DA neuron degeneration (Good et al., FASEB J 2011, PMID: 21233488). We also acknowledge that alterations to dopamine neuron activity are likely complex in PD, and that dopamine neuron health and function can be impacted not just by simple increases in activity, but also by changes in activity patterns and regularity. We have amended our discussion to include the important caveat of changes in activity occurring as compensation, as well as further evidence of changes in activity preceding dopamine neuron death.

There is a very large, multi-decade literature on calcium elevation and its effects on neuronal loss in many different types of neurons. The authors should discuss their findings in this context and refer to some of this previous work. In a nutshell, the observations of the present manuscript could be summarized by stating that the chronic membrane depolarization induced by the CNO treatment is likely to induce a chronic elevation of intracellular calcium and this is then likely to activate some of the well-known calcium-dependent cell death mechanisms. Whether such cell death is linked in any way to PD is not really demonstrated by the present results. The authors are encouraged to perform a thorough revision of the discussion to address all of these issues, discuss the major limitations of the present model, and refer to the broad pre-existing literature linking membrane depolarization, calcium, and neuronal loss in many neuronal cell types.

While our model demonstrates classic excitotoxic cell death pathways, we would like to emphasize both the chronic nature of our manipulation and the progressive changes observed, with increasing degeneration seen at 1, 2, and 4 weeks of hyperactivity in an axon-first manner. This is a unique aspect of our study, in contrast to much of the previous literature which has focused on shorter timescales. Thus, while we have revised the discussion to more comprehensively acknowledge previous studies of calcium-dependent neuron cell death, we believe we have made several new contributions that are not predicted by existing literature. We have shown that this chronic manipulation is specifically toxic to nigral dopamine neurons, and the data that VTA dopamine neurons continue to be resilient even at 4 weeks is interesting and disease-relevant. We therefore do not want to use findings

from other neuron types to draw assumptions about DA neurons, which are a unique and very diverse population. We acknowledge that as with all preclinical models of PD, we cannot draw definitive conclusions about PD with this data. However, we reiterate that we strongly believe that drawing connections to human disease is important, as dopamine neuron activity is very likely altered in PD and a clearer understanding of how dopamine neuron survival is impacted by activity will provide insight into the mechanisms of PD.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

(1) The temporal design of the experiments is quite confusing. For instance, Figures 1 and 3 illustrate the daily changes of the mice and suggest some critical time points within 2 weeks of CNO administration, whereas Figure 2 presents data at 2 and 4 weeks, which are much later than the proposed critical time points. Furthermore, Figure 4 includes only 1 week data, and lacks subsequent data from 2 and 4 weeks, at which significant changes such as calcium levels and neuronal/axonal degeneration are observed.

While interesting behavior and calcium phenotypes were detected within 2 and 4 weeks of CNO administration (Figures 1 and 3), we only collected tissues for histology at the 2 and 4 week time points (Figure 2). Observing degeneration of DA neuron axons but not cell bodies at 2 weeks served as a rationale to extend to the 4 week time point to determine whether degeneration was progressive. At the same time, our primary focus is on identifying early changes that may drive or contribute to the degeneration. As such, we recorded calcium changes over a 2-week treatment period, capturing the period during which almost all of the dopamine axons are lost. Similarly, we had the capacity to perform spatial transcriptomics at only one time point, and the 1 week time point was selected to capture transcriptomic changes that precede and potentially contribute to the mild and severe degeneration that occurs at 2 and 4 weeks, respectively. We have added text clarifying the rationale for the time points chosen.

(2) The authors showed the changes in neuronal firing in dopamine neurons by the administration of CNO. However, one of the most important features of dopaminergic neuronal activity is dopamine release at its axon terminals in the striatum. Thus, the claims raised in this paper would be better supported if the authors further show any alterations in dopamine release (by FSCV or fluorescent dopamine sensors) at some critical time points during or after CNO application.

While we are confident that DA release is altered due to the significant changes in behavior when hM3Dq DREADDs are activated specifically in DA neurons, the current manuscript does not quantify this, or distinguish between axonal and somatodendritic DA release. Interestingly, we did find significantly decreased striatal dopamine by HPLC after chronic activation (Figure S6). We believe that resolving these questions is beyond the scope of this manuscript, but have added text indicating the importance of these experiments.

(3) The authors used 2% sucrose as a vehicle via drinking water. Please explain the rationale behind this choice.

We used 2% sucrose as the vehicle because it is also added to the CNO water to counteract the bitterness of CNO (Kumar et al., J Neurotrauma 2024, PMID: 37905504). We have clarified this in the manuscript.

(4) As we know, mRNA levels of some genes do not always predict their protein levels; there is sometimes a huge discrepancy between mRNA and protein abundance. In this paper, the mechanistic interpretation of the results by the authors heavily relies on the

spatial transcriptomics of the midbrain and striatum. Thus, the authors need to provide additional data proving that the gene expression of some genes in the CNO group is also changed at the level of protein.

We agree that validating hits at the protein level is valuable, however we were limited in our ability to assess these changes for the revision. However, we have done additional transcriptomics with the high resolution Xenium platform to increase confidence in a subset of hits of interest for follow up in future work, and we included data on genes related to DA metabolism and markers of DA neurons.

(5) The authors provided spatial transcriptomics data only for mice with one week of chronic activation. However, other data also indicate significant differences when the activation period extends beyond 10 to 12 days (Figure 1C, Figure 3D-F). While a 7-day chronic activation time point might be crucial, additional transcriptomics data from later time points would be beneficial to confirm the persistence of these changes in gene expression. Furthermore, differential gene expression (DEG) analysis at these later time points could identify novel pathways or genes influenced by the chronic activation of dopamine neurons.

This is an interesting point and would provide valuable data as to how chronic activity influences gene expression, however additional transcriptomics at later timepoints is beyond the scope of this paper. In future studies we will assess changes observed in this manuscript at other time points.

(6) Figure 1D, Figure S1C:

The authors should present the sample recording traces to demonstrate that the electrophysiological recordings were appropriately made.

These data have been provided in Figure S2.

(7) Figure S1C:

AP thresholds in SNc dopamine neurons from both groups look quite high. In addition, considering the data from the previous reports, AP peak amplitudes in SNc dopamine neurons from both groups seem to be very low. Are these values correct?

The thresholds and peaks are correct, including the AP (threshold to peak), which is typical in our (Dr. Margolis's) experience. AP thresholds are measured from an average of at least 10 APs, as the voltage at which the derivative of the trace first exceeds 10 V/s. As mentioned in the methods section, junction potentials were not corrected, which can result in values that are a bit depolarized from ground truth. This junction potential would be consistent across all recordings, thus not impede detection of a difference in AP thresholds between groups of animals.

(8) Figure 1E:

It would be better if the statistical significance is depicted in the graph.

We don't perform repeated measures statistics across data like these, as the data are continuous, collected at 10 kHz. For ease of displaying the data, the data for each neuron is binned and then these traces are averaged together. We display SEM to give a sense of the variance across neurons. We have provided sample traces of individual neurons to better demonstrate the variability and significance of this data (Figure S2).

The representative staining images appear to be taken from coronal slices at anatomically different positions along the rostral-to-caudal axis. Although the total numbers of TH+ cells are comparable between vehicle and CNO groups in the graph, the sample images do not reflect this result. The authors should replace the current images with the better ones.

We have replaced this image in the manuscript.

Reviewer #2 (Recommendations For The Authors):

Minor concerns:

(1) The authors claim that their transcriptomics experiments are conducted 'before any degeneration has occurred'. And they do not see significant differences in the TH expression in the striatum. However, the n for these mice at 1 week is lower than the n use at 2 weeks (n=5 vs n=8-9) and the images used to show 'no degeneration' really look like there is some degeneration going on. Also, throughout the paper, there is a stronger effect when degeneration is measured with mCherry compared to when it is measured with TH. The 'no change' claim is made only with the TH comparison. It seems possible (and almost likely) that there would be significant axonal degeneration at one week with either a higher sample size or using the mCherry comparison. The authors should simply claim that their transcriptomics data is collected before any 'somatic' degeneration occurs.

Thank you, we have included data that shows partial terminal loss after one week of activation (Figure S3B, Figure S5A) and have corrected this language in the manuscript to reflect transcriptomics occurring before somatic degeneration.

(2) While selective degeneration is one of the most interesting findings in the paper, that finding is not emphasized and why it would be interesting to compare the VTA vs SNc is not discussed in the introduction.

Emphasis for comparing the VTA vs the SNc has been added to the introduction, along with additional electrophysiology data in VTA dopamine neurons in Figure 1 and Figure S2.

(3) In a similar direction, the vulnerability of dopaminergic neurons has been shown to be differential even within the SNc, with the ventral tier neurons degenerating more severely and the dorsal tier neurons remaining resilient. Is there any evidence for a ventral-dorsal degeneration gradient in the SNc in these experiments?

This is a really interesting point and changes to dopamine neuron subtypes along the ventral-dorsal axis may be occurring in this model, particularly as there is more selective loss of SNc neurons. However, the cell type involved would be difficult to determine at this stage, since single cell transcriptomic resolution is necessary across the entire SNc to identify cell subtypes. Transcriptomic identification is further complicated given that transcriptome change has recently been shown with genetic manipulation (Gaertner *et al.*, bioRxiv 2024, PMID: 38895448), and we would think could similarly change with increased activity. Assessing these issues are beyond the scope of this paper.

(4) The running data is very interesting and the circadian rhythm alterations are compelling.

However, it is unclear whether the CNO mice run more total compared with the vehicle mice.

The authors should show the combined total running data to evaluate this. We now show total running data in Figure 1C.

(5) The finding that acute CNO has no effect on the membrane potential of SNc neurons after chronic CNO exposure is very peculiar! Especially because the fiber photometry data suggests that CNO continues to have an effect in vivo. Is there any explanation for this?

While there is no acute electrophysiological response to CNO detected in this group, there may be intracellular pathways activated by the DREADD that do not acutely impact membrane potential in current clamp ($I = 0$ pA) mode.

(6) The terminology of chronic CNO is sometimes confusing as it refers to both 2-week and 4week administration. Using additional terminology such as 'early' and 'late' might help with clarity.

We have decreased usage of 'chronic,' and increased usage of more specific treatment times in order to increase clarity throughout the manuscript.

(7) In Figure 2C, the SNc image looks binarized.

This image has been updated.

(8) Also in Figure 2, why are TH and mCherry measured for the 4-week time point, but only TH measured for the 2-week time point?

mCherry quantification was performed to further support the finding of DA neuron death, and was therefore not assessed at 2 weeks given that there was no change in the TH stereology.

(9) Additional scale bars and labeling is needed in Figure 3. In addition, there is such a strong reduction in noise after chronic CNO in the fiber photometry recordings, and the noise does not return upon CNO washout. What is the explanation for this?

Additional scale bars were added to Figure 3. Traces are not getting less noisy with chronic CNO treatment, rather, there is less bursting activity in the dopamine cells. Our interpretation is that the baseline activity is rescued during washout but this bursting activity is not.

(10) While not necessary to support the claims in this paper, it would be very interesting to see if chronic inhibition of dopaminergic neurons had a similar or different effect, as too little dopaminergic activity may also cause degeneration in some cases.

We agree that assessing chronic inhibition is valuable, and this is an important area for future research.

Reviewer #3 (Recommendations For The Authors):

All the mice used in the study are not listed in the methods section. For example, the GCaMP6f floxed mice discussed in the results section are not listed in the methods. Also, the breeding scheme used for the different mouse lines needs to be described. For example, did the DAT-Cre mice carry one or two alleles?

Both the DAT^{ires}Cre and GCaMP6f floxed (Ai148) Jax mouse line numbers and RRIDs are included in the methods. DAT^{ires}Cre mice carried two alleles.

| *In the methods section, the amount of virus injected needs to be mentioned.*

This information has been added to the methods section.

| *In all result graphs, please include the individual data points so that the readers can see the distribution of the data and quickly see the sample size.*

Graphs have been updated to include all individual data points. For line graphs, the distribution is communicated by the error bars, while the n is in the legends.

| *The authors provide running wheel data in supplementary figure 1A to validate that chemogenetic activation of dopamine neurons leads to increased locomotor activity. The results shown in the figure appear to be qualitative as no average data is presented. The authors should provide average data from all mice tested.*

Average IP response data for all mice assessed for running wheel activity has been included in Figure S1.

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